Part I

Basic principles

1

Properties of foods and processing theory

1.1 Properties of liquids, solids and gases

Liquids, gases and some solids (for example powders and particulate materials) are termed 'fluids' and can flow without disintegration when a pressure is applied to them. In contrast, solids deform when pressure is applied to them (Section 1.1.4). In this section, the properties of fluids and solids that are relevant to both the design of food processes and the quality of processed food are described. More detailed mathematical treatments and derivations of formulae used in food engineering calculations are given in a number of texts including Toledo (1999), Lewis (1990), Brennan *et al*. (1990) and Earle (1983).

The transition from solid to fluid and back is known as a *phase transition* and this is important in many types of food processing (e.g. water to water vapour in evaporation and distillation (Chapter 13) and dehydration (Chapter 15); water to ice in freezing (Chapter 21) and freeze drying or freeze concentration (Chapter 22) or crystallisation of fats (Chapter 23)). Phase transition takes place isothermally at the phase transition temperature by release or absorption of latent heat, and can be represented by a *phase diagram* (for example Chapter 22, Fig. 22.1).

A second type of transition, known as *glass transition*, takes place without the release or absorption of latent heat and involves the transition of a food to an amorphous glass state at its glass transition temperature. The transition is dependent on the temperature of the food, time, and the moisture content of the food. Examples of glass transition temperatures are given in Chapter 21 (Table 21.2). When materials change to glasses, they do not become crystalline, but retain the disorder of the liquid state. However, their physical, mechanical, electrical and thermal properties change as they undergo the transition. In their glassy state, foods become very stable because compounds that are involved in chemical reactions that lead to deterioration are immobilised and take long periods of time to diffuse through the material to react together. Details of nine key concepts underlying the relationship between molecular mobility and food stability are given by Fennema (1996). Processes that are significantly influenced by transition to a glassy state include aroma retention, crystallisation, enzyme activity, microbial activity, non-enzymic browning, oxidation, agglomeration and caking. The relationship between

glass transition and water activity (Section 1.5) and factors that affect glass transition are described in detail by Rahman (1999), Blanshard (1995) and Fennema (1996).

1.1.1 Density and specific gravity

A knowledge of the density of foods is important in separation processes (Chapter 6), and differences in density can have important effects on the operation of size reduction and mixing equipment (Chapters 4 and 5). The density of a material is equal to its mass divided by its volume and has units of $kg\,m^{-3}$. Examples of the density of solid foods and other materials used in food processing are shown in Table 1.1 (see also Chapter 15, Table 15.7) and examples of densities of liquids are shown in Table 1.2. The density of materials is not constant and changes with temperature (higher temperatures reduce the density of materials) and pressure. This is particularly important in fluids where differences in density cause convection currents to be established (Section 1.4.3).

Table 1.1 Densities of foods and other materials

Material	Density ($kg\,m^{-3}$)	Bulk density ($kg\,m^{-3}$)	Temperature (°C)
Solids			
Aluminium	2640	–	0
Copper	8900	–	0
Stainless steel	7950	–	20
Concrete	2000	–	20
Grapes	1067	368	–
Tomatoes	–	672	–
Lemons/oranges	–	768	–
Fresh fruit	865–1067	–	–
Frozen fruit	625–801	–	–
Fresh fish	967	–	–
Frozen fish	1056	–	–
Water (0°C)	1000	–	0
Ice (0°C)	916	–	0
Ice (−10°C)	933	–	−10
Ice (−20°C)	948	–	−20
Fat	900–950	–	20
Salt	2160	960	–
Sugar (granulated)	1590	800	–
Sugar (powdered)	–	480	–
Starch	1500	–	–
Wheat	1409–1430	790–819	
Wheat flour	–	480	–
Barley	1374–1415	564–650	–
Oats	1350–1378	358–511	–
Rice	1358–1386	561–591	–
Gases			
Air	1.29	–	0
Air	0.94	–	100
Carbon dioxide	1.98	–	0
Carbon dioxide	1.46	–	100
Nitrogen	1.30	–	0

Adapted from data of Earle (1983), Lewis (1990), Milson and Kirk (1980), Peleg (1983) and Mohsenin (1970).

Table 1.2 Properties of fluids

	Thermal conductivity ($W\,m^{-1}\,{}^{o}K^{-1}$)	Specific heat ($kJ\,kg^{-1}\,{}^{o}K^{-1}$)	Density ($kg\,m^{-3}$)	Dynamic viscosity ($N\,s\,m^{-2}$)	Temperature (^{o}C)
Air	0.024	1.005	1.29	1.73×10^{-5}	0
	0.031	1.005	0.94	2.21×10^{-5}	100
Carbon dioxide	0.015	0.80	1.98		0
Oxygen		0.92		1.48×10^{-3}	20
Nitrogen	0.024	1.05	1.30		0
Refrigerant 12	0.0083	0.92			
Water	0.57	4.21	1000	1.79×10^{-3}	0
	0.68	4.21	958	0.28×10^{-3}	100
Sucrose solution (60%)				6.02×10^{-2}	20
Sucrose solution (20%)	0.54	3.8	1070	1.92×10^{-3}	20
Sodium chloride solution (22%)	0.54	3.4	1240	2.7×10^{-3}	2
Acetic acid	0.17	2.2	1050	1.2×10^{-3}	20
Ethanol	0.18	2.3	790	1.2×10^{-3}	20
Rape-seed oil			900	1.18×10^{-1}	20
Maize oil		1.73			20
Olive oil	0.168			8.4×10^{-2}	29
Sunflower oil		1.93			20
Whole milk	0.56	3.9	1030	2.12×10^{-3}	20
				2.8×10^{-3}	10
Skim milk			1040	1.4×10^{-3}	25
Cream (20% fat)			1010	6.2×10^{-3}	3
Locust bean gum (1% solution)				1.5×10^{-2}	
Xanthan gum (1% solution)			1000		

From Earle (1983), Lewis (1987) and Peleg and Bagley (1983).

The density of liquids is a straightforward measure of mass/volume at a particular temperature, but for particulate solids and powders there are two forms of density: the density of the individual pieces and the density of the bulk of material, which also includes the air spaces between the pieces. This latter measure is termed the *bulk density* and is 'the mass of solids divided by the bulk volume'. The fraction of the volume that is taken up by air is termed the *porosity* (ϵ) and is calculated by:

$$\epsilon = V_a/V_b \qquad 1.1$$

where V_a = volume of air (m^3) and V_b = volume of bulk sample (m^3).

The bulk density of a material depends on the solids density and the geometry, size and surface properties of the individual particles. Examples of bulk densities of foods are shown in Table 1.1 and bulk density is discussed in relation to spray dried powders in Chapter 15.

The density of liquids can be expressed as *specific gravity* (SG), a dimensionless number (Appendix D), which is found by dividing the mass (or density) of a liquid by the mass (or density) of an equal volume of pure water at the same temperature:

$$\text{SG} = \text{mass of liquid/mass water} \qquad 1.2$$

$$\text{SG} = \text{density of liquid/density water} \quad [1.3]$$

If the specific gravity of a liquid is known at a particular temperature, its density can be found using:

$$\rho_L = (SG)_T \cdot \rho_w \quad [1.4]$$

where ρ_L = liquid density (kg m^{-3}) and ρ_w = density of water, each at temperature T (^oC).

Specific gravity is widely used instead of density in brewing and other alcoholic fermentations (Chapter 7), where the term 'original gravity (OG)' is used to indicate the specific gravity of the liquor before fermentation (for example '1072' or '72' refers to a specific gravity of 1.072).

The density of gases depends on their pressure and temperature (Table 1.1). Pressure is often expressed as 'gauge pressure' when it is above atmospheric pressure, or as 'gauge vacuum' when it is below atmospheric pressure. Pressure is calculated using the *Ideal Gas Equation* as follows:

$$PV = nRT \quad [1.5]$$

where P = absolute pressure (Pa), V = volume (m^3), n = number of moles of gas (kmole), R = the gas constant (8314 N(m)/(kmole K) and T = temperature (K).

This equation is useful for calculation of gas transfer in applications such as modified atmosphere storage or packaging (Chapter 20), cryogenic freezing (Chapter 21) and the permeability of packaging materials (Chapter 24).

Sample problem 1.1
Calculate the amount of oxygen that enters through a polyethylene packaging material in 24 h if the pack has a surface area of 750 cm^2 and an oxygen permeability of 120 ml m^{-2} per 24 h at 23 ^oC and 85% relative humidity (see Chapter 24, Table 24.2).

Solution to Sample problem 1.1
The volume of oxygen entering through the polyethylene:

$$V = 120 \times \frac{750}{100^2}$$

$$= 9.0 \text{ cm}^3$$

using equation 1.5,

$$n = 9.0 \times 1000/(8314 \times 273)$$

$$= 0.00396 \text{ kmole}$$

The density of gases and vapours is also referred to as *specific volume* which is 'the volume occupied by unit mass of gas or vapour' and is the inverse of density. This is used for example in the calculation of the amount of vapour that must be handled by fans during dehydration (Chapter 15) or by vacuum pumps in freeze drying (Chapter 22) or vacuum evaporation (Chapter 13). Further details are given by Lewis (1990) and Toledo (1999).

When air is incorporated into liquids (for example cake batters, ice cream, whipped cream) it creates a foam and the density is reduced. The amount of air that is incorporated

is referred to as the *over-run* and is described in Chapter 4 (Section 4.2.3)

$$\text{over-run} = \frac{\text{volume of foam} - \text{volume of liquid}}{\text{volume of liquid}} \times 100$$

Typical over-run values are 60–100% for ice cream and 100–120% for whipped cream.

1.1.2 Viscosity

Viscosity is an important characteristic of liquid foods in many areas of food processing. For example the characteristic mouth feel of food products such as tomato ketchup, cream, syrup and yoghurt is dependent on their consistency or viscosity.[1] The viscosity of many liquids changes during heating, cooling, concentration, etc. and this has important effects on, for example, the power needed to pump these products. Viscosities of some common fluids in food processing are shown in Table 1.2.

Viscosity may be thought of as a liquid's internal resistance to flow. A liquid can be envisaged as having a series of layers and when it flows over a surface, the uppermost layer flows fastest and drags the next layer along at a slightly lower velocity, and so on through the layers until the one next to the surface is stationary. The force that moves the liquid is known as the *shearing force* or *shear stress* and the velocity gradient is known as the *shear rate*. If shear stress is plotted against shear rate, most simple liquids and gases show a linear relationship (line A in Fig. 1.1) and these are termed *Newtonian* fluids. Examples include water, most oils, gases, and simple solutions of sugars and salts. Where the relationship is non-linear (lines B–E in Fig. 1.1), the fluids are termed 'non-Newtonian'. Further details are given by Nedderman (1997). For all liquids, viscosity decreases with an increase in temperature but for most gases it increases with temperature (Lewis, 1990).

Many liquid foods are non-Newtonian, including emulsions and suspensions, and concentrated solutions that contain starches, pectins, gums and proteins. These liquids often display Newtonian properties at low concentrations but as the concentration of the solution is increased, the viscosity increases rapidly and there is a transition to non-Newtonian properties (Rielly, 1997). Non-Newtonian fluids can be classified broadly into the following types:

- pseudoplastic fluid (line B in Fig. 1.1) – viscosity decreases as the shear rate increases (e.g. emulsions, and suspensions such as concentrated fruit juices and purées)
- dilatant fluid (line C in Fig. 1.1) – viscosity increases as the shear rate increases. (This behaviour is less common but is found with liquid chocolate and cornflour suspension.)
- Bingham or Casson plastic fluids (lines D and E in Fig. 1.1) – there is no flow until a critical shear stress is reached and then shear rate is either linear (Bingham type) or non-linear (Casson type) (e.g. tomato ketchup)
- thixotropic fluid – the structure breaks down and viscosity decreases with continued shear stress (most creams)
- rheopectic fluid – the structure builds up and viscosity increases with continued shear stress (e.g. whipping cream)

1. Although these two terms are often used inter-changeably, strictly 'viscosity' is only applied to Newtonian fluids – see below.

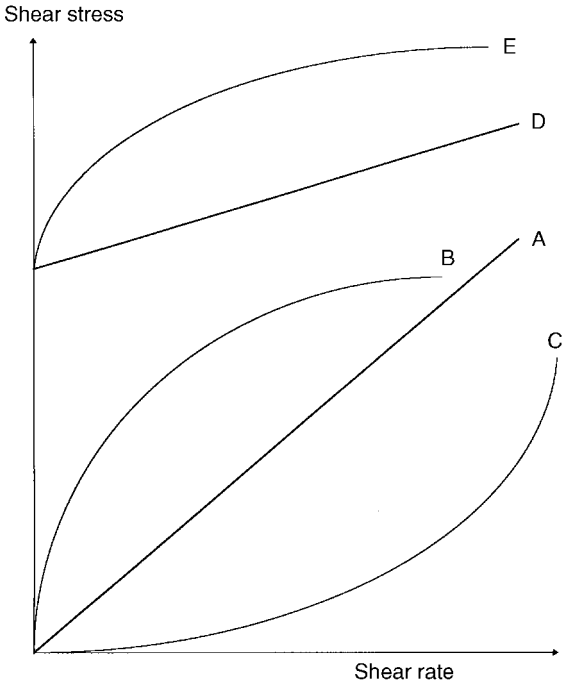


Fig. 1.1 Changes in viscosity of Newtonian fluid (A) and different types of non-Newtonian fluids; (B) pseudoplastic fluid; (C) dilatant fluid; (D) Bingham plastic fluid and (E) Casson plastic fluid. (After Lewis (1990).)

- viscoelastic material – has viscous and elastic properties exhibited at the same time. When a shear stress is removed the material never fully returns to its original shape and there is a permanent deformation (e.g. dough, cheese, gelled foods).

The measurement of viscosity is complicated by the range of terms used to describe it. The simplest is the ratio of shear stress to shear rate, which is termed the *dynamic viscosity* ($kg\,m^{-1}\,s^{-1}$). This is related to another term, *kinematic viscosity* ($m^2\,s^{-1}$), as follows:

$$\text{kinematic viscosity} = \frac{\text{dynamic viscosity}}{\text{density}} \qquad 1.6$$

Other terms, including 'relative viscosity', 'specific viscosity' and 'apparent viscosity', together with descriptions of methods of measuring viscosity, are given by Lewis (1990) and Toledo (1999).

1.1.3 Surface activity

A large number of foods comprise two or more immiscible components, which have a boundary between the phases (Table 1.3). The phases are known as the *dispersed phase* (the one containing small droplets or particles) and the *continuous phase* (the phase in which the droplets or particles are distributed). Details of the preparation of emulsions are given in Chapter 4.

One characteristic of these systems is the very large surface area of the dispersed phase that is in contact with the continuous phase. In order to create the increased surface

Table 1.3 Examples of colloidal food systems

Dispersed phase	Continuous phase	Name	Example
Liquid	Gas	Fog, mist, aerosol	Sprays (e.g. spray drying)
Solid	Gas	Smoke, aerosol	Carbon dioxide fog, smoke
Gas	Liquid	Foam	Whipped cream, aeration of fermentation media, hydrogenation of oils (hydrogen in oil)
Liquid	Liquid	Emulsion	Cream, mayonnaise, margarine and low-fat spreads, salad cream, sausagemeat and cakes
Solid	Liquid	Sol, colloidal solution, gel suspension	Chocolate drinks, fruit juice
Gas	Solid	Solid foam	Meringue, ice cream

From Lewis (1990).

area, a considerable amount of energy needs to be put into the system using for example a high-speed mixer or an homogeniser. Droplets are formed when new surfaces are created. To understand the reason for this it is necessary to know the forces acting in liquids: within the bulk of a liquid the forces acting on each individual molecule are equal in all directions and they cancel each other out. However, at the surface the net attraction is towards the bulk of the liquid and as a result, the surface molecules are 'pulled inwards' and are therefore in a state of tension (produced by surface tension forces). This causes liquid droplets to form into spheres because this shape has the minimum surface area for the particular volume of liquid.

Emulsions

Chemicals that reduce the surface tension in the surface of a liquid are termed *surface active* and are known as 'surfactants', 'emulsifying agents' or 'detergents'. By reducing the surface tension, they permit new surfaces to be produced more easily when energy is put into the system (for example by homogenisers) and thus enable larger numbers of droplets to be formed. There are naturally occurring surfactants in foods, including alcohols, phospholipids and proteins and these are sometimes used to create food emulsions (for example using egg in cake batters). However, synthetic chemicals have more powerful surface activity and are used in very small amounts to create emulsions. Others are used in detergents for cleaning operations (Chapter 26). Details of synthetic emulsifying agents are given in Chapter 4 and Appendix C.

Surface active agents contain molecules which are polar (or 'hydrophilic') at one end and non-polar (or 'lipophilic') at the other end. In emulsions, the molecules of emulsifying agents become oriented at the surfaces of droplets, with the polar end in the aqueous phase and the non-polar end in the oil phase.

Detergents are surface active agents that reduce the surface tension of liquids to both promote wetting (spreading of the liquid) and to act as emulsifying agents to dissolve fats. The detergent molecules have a lipophilic region of long chain fatty acids and a hydrophilic region of either a sodium salt of carboxylic acid (soapy detergents) or the sodium salt of an alkyl or aryl sulphonate (anionic detergents). Anionic detergents are not affected by hard water, whereas soapy detergents form a scum in hard water. Non-ionic detergents, which have alcohols, esters or ethers as the hydrophilic component, produce little foam and are easily rinsed off. Enzymes may also be added to detergents to remove proteins, and other ingredients may include polyphosphates (to soften water and keep dirt

in suspension), sodium sulphate or sodium silicate (to make detergent powder free-flowing) and sodium perborate (bleaching agent).

Foams

Foams are two-phase systems which have gas bubbles dispersed in a liquid or a solid, separated from each other by a thin film. In addition to food foams (Table 1.3), foams are widely used for cleaning equipment. The main factors needed to produce a stable foam are:

- a low surface tension to allow the bubbles to contain more air and prevent them contracting
- gelation or insolubilisation of the bubble film to minimise loss of the trapped gas and to increase the rigidity of the foam and
- a low vapour pressure in the bubbles to reduce evaporation and rupturing of the film.

In food foams, the structure of the foam may be stabilised by freezing (ice cream), by gelation (setting gelatin in marshmallow), by heating (cakes, meringues) or by the addition of stabilisers such as proteins or gums (Appendix C).

1.1.4 Rheology and texture

The texture of foods has a substantial influence on consumers' perception of 'quality' and during chewing (or 'mastication'), information on the changes in texture of a food is transmitted to the brain from sensors in the mouth, from the sense of hearing and from memory, to build up an image of the textural properties of the food. This may be seen as taking place in a number of stages:

1. an initial assessment of hardness, ability to fracture and consistency during the first bite
2. a perception of chewiness, adhesiveness and gumminess during chewing, the moistness and greasiness of the food, together with an assessment of the size and geometry of individual pieces of food
3. a perception of the rate at which the food breaks down while chewing, the types of pieces formed, the release or absorption of moisture and any coating of the mouth or tongue with food.

These various characteristics have been categorised (Table 1.4) and used to assess and monitor the changes in texture that affect the quality of foods.

Rheology is the science of deformation of objects under the influence of applied forces. When a material is stressed it deforms, and the rate and type of deformation characterise its rheological properties. (The rheological properties of solid foods are described in more detail in Chapter 4.) A large number of different methods have been used to assess the texture of food, including texture profiling by sensory methods using taste panels (e.g. Bourne, 1982), Quantitative Descriptive Analysis (QDA) (Fig. 1.2), described by Clark (1990), and empirical methods in which measurements of the forces needed to shear, penetrate, extrude, compress or cut a food are related to a textural characteristic. These methods are reviewed by Kilcast (1999), Rosenthal (1999), Lawless and Heyman (1998) and Brennan (1984).

Examples of these instrumental methods include the Brabender system to measure dough texture or the viscosity of starch pastes, cone or rod penetrometers to measure the yield stress of margarine and spreads or the hardness of fruits, the General Foods Texturometer which simulates mastication by compressing foods using a plunger, and the

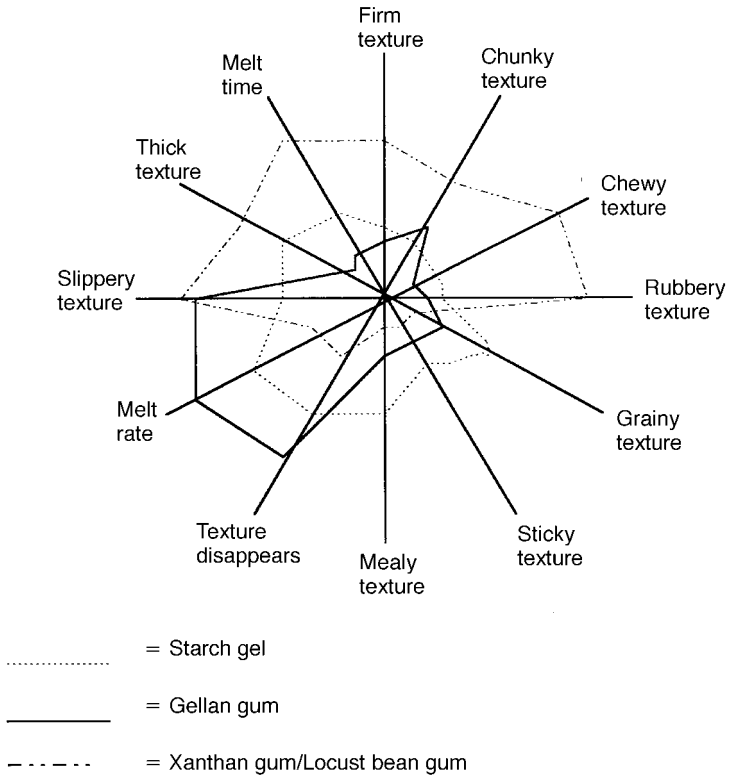


Fig. 1.2 Example of texture assessment using quantitative descriptive analysis. (Adapted from Clark (1990).)

Table 1.4 Textural characteristics of foods

Primary characteristic	Secondary characteristic	Popular terms
Mechanical characteristics		
Hardness		Soft→firm→hard
Cohesiveness	Brittleness	Crumbly, crunchy, brittle
	Chewiness	Tender, chewy, tough
	Gumminess	Short, mealy, pasty, gummy
Viscosity		Thin, viscous
Elasticity		Plastic, elastic
Adhesiveness		Sticky, tacky, gooey
Geometrical characteristics		
Particle size and shape		Gritty, grainy, coarse
Particle shape and orientation		Fibrous, cellular, crystalline
Other characteristics		
Moisture content		Dry→moist→wet→watery
Fat content	Oiliness	Oily
	Greasiness	Greasy

Adapted from Szczesniak (1963).

Instron Universal Testing machine, which measures stress and strain forces by compression or extension. Chemical methods include measurement of starch or pectin content, and microscopic methods include electron microscopy of emulsions or the flesh structure of meat and fish. These methods are described in detail by Sherman (1979), Prentice (1984), Bourne (1978), Brennan (1984), Kramer and Szczesniak (1973), Lewis (1990) and Bourne (1982).

1.2 Material transfer

The transfer of matter is an important aspect of a large number of food processing operations: it is a key factor in solvent extraction, distillation (Chapter 13) and membrane processing (Chapter 6) and it is an important factor in loss of nutrients during blanching (Chapter 10). Mass transfer of gases and vapours is a primary factor in evaporation (Chapter 13), dehydration (Chapter 15), baking and roasting (Chapter 16), frying (Chapter 17), freeze drying (Chapter 22), the cause of freezer burn during freezing (Chapter 21) and a cause of loss in food quality in chilled, MAP and packaged foods (Chapters 19, 20 and 24, respectively).

In an analogous way to heat transfer (Section 1.4), the two factors that influence the rate of mass transfer are a driving force to move materials and a resistance to their flow. When considering dissolved solids in liquids, the driving force is a difference in the solids concentration, whereas for gases and vapours, it is a difference in partial pressure or vapour pressure. The resistance arises from the medium through which the liquid, gas or vapour moves and any interactions between the medium and the material.

An example of materials transfer is diffusion of water vapour through a boundary layer of air in operations such as dehydration, baking, etc. Packaging also creates additional boundary layers which act as barriers to movement of moisture and to heat transfer (Fig. 1.3).

The rate of diffusion is found using:

$$N_A = D_w/RT_x \,.\, P_T/P_{Am} \,.\, (P_{w1} - P_{w2}) \qquad \boxed{1.7}$$

where N_A (kg s^{-1}) or (kmols s^{-1}) = rate of diffusion, D_w = diffusion coefficient of water vapour in air, R = the Gas Constant (= 8.314 kJ $kmol^{-1}$ K^{-1}), T (K) = temperature, x (m) = distance across stationary layer, P_T (kNm^{-2}) = total pressure, P_{Am} (kNm^{-2}) = mean pressure of non-diffusing gas across the stationary layer and $P_{w1} - P_{w2}$ (kNm^{-2}) = water vapour pressure driving force.

Formulae for diffusion of solutes through liquids and for gases dissolving in liquids are given by Lewis (1990) and Toledo (1999).

Mass balances

The law of conversion of mass states that 'the mass of material entering a process equals the mass of material leaving'. This has applications in, for example, mixing (Chapter 5), fermentation (Chapter 7), and evaporation (Chapter 13).

In general a mass balance for a process takes the following form:

mass of raw materials in = mass of products and wastes out + mass of stored materials + losses

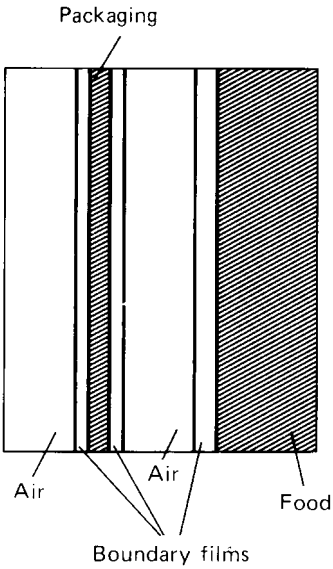


Fig. 1.3 Barriers to mass transfer and heat flow due to packaging.

Many mass balances are analysed under steady-state conditions where the mass of stored materials and losses are equal to zero. Mass balances are used to calculate the quantities of materials in different process streams, to design processes, to calculate recipe formulations, the composition after blending, process yields and separation efficiencies. A typical mass balance is shown in Fig. 1.4.

Here the total mass balance is

$$W + A = \text{moist air} + D$$

The mass balance for air is

$$A + \text{moisture} = \text{moist air}$$

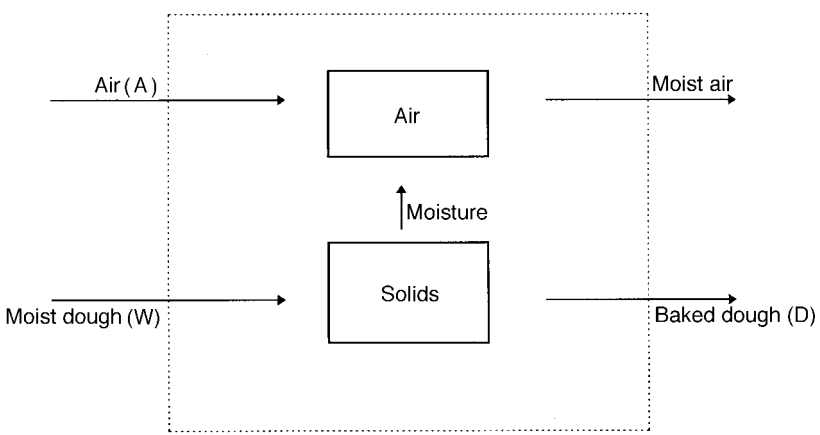


Fig. 1.4 Diagram of material flow during baking in an oven.

The mass balance for solids is

$$W = \text{moist air} + D$$

Examples of mass balances calculations are shown in Chapter 13 (Sample problem 13.1) and Chapter 15 (Sample problem 15.2).

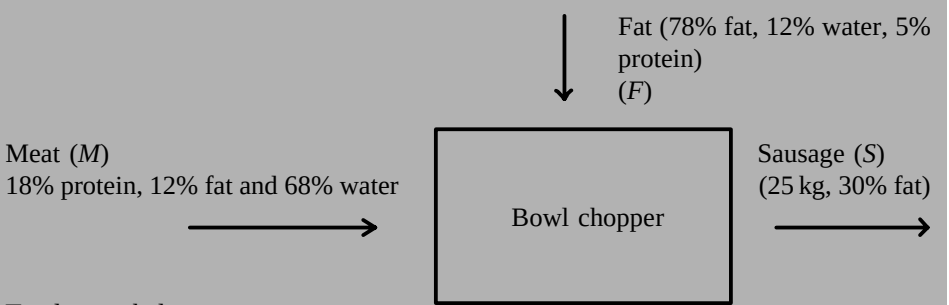
In applications involving concentration or dilution, the use of mass fraction or weight percentage is often used:

$$\text{Mass fraction A} = \frac{\text{mass of component A}}{\text{total mass of mixture}}$$

or

Sample problem 1.2
Calculate the total mass balance and component mass balance for mixing ingredients to make 25 kg of beef sausages having a fat content of 30%, using fresh beef meat and beef fat. Typically, beef meat contains 18% protein, 12% fat and 68% water and beef fat contains 78% fat, 12% water and 5% protein.

Solution to Sample problem 1.2
Let F = mass of beef fat (kg)
Let M = mass of beef meat (kg)



Total mass balance

$$M + F = 25$$

Fat balance

$$0.12M + 0.78F = 0.3(25)$$

Substitute $M = 25 - F$ into the fat balance

$$0.12(25 - F) + 0.78F = 7.5$$
$$3.0 - 0.12F + 0.78F = 7.5$$
$$= 6.82 \text{ kg}$$

and

$$M = 25 - 6.82$$
$$= 18.18 \text{ kg}$$

$$\text{Total mass of mixture} = \frac{\text{mass of component A}}{\text{mass fraction of A}}$$

If the weight of the component and its mass fraction are known, the total mass of the mixture can be calculated. Computerised recipe calculations are described in Chapter 2 (Section 2.2.1).

A simple method to calculate the relative masses of two materials that are required to form a mixture of known composition is the *Pearson Square* (Anon., 1996). If, for example, homogenised milk (3.5% fat) is to be mixed with cream (20% fat) to produce a light cream containing 10% fat, the Pearson Square (Fig. 1.5) is constructed with the fat composition of ingredients on the left side, the fat content of the product in the centre. By subtracting diagonally across the square, the resulting proportions of milk and cream can be found (i.e. 10 parts milk and 6.5 parts cream in Fig 1.5). An alternative method is to use a total balance and a fat balance (see Sample problem 1.2). Toledo (1999) describes the application of mass balances to multi-stage processing and includes a computer programme for the use of mass balances in recipe formulations. Another example of the use of a Pearson Square is shown in Sample problem 1.3.

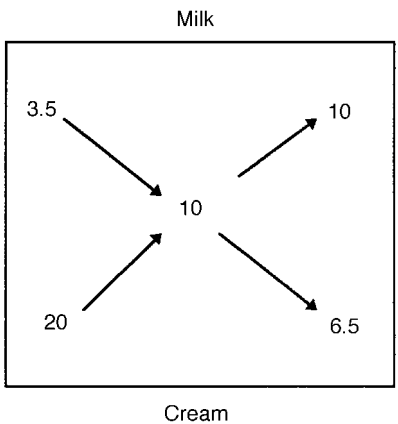


Fig. 1.5 Pearson Square.
(From Anon. (1996).)

1.3 Fluid flow

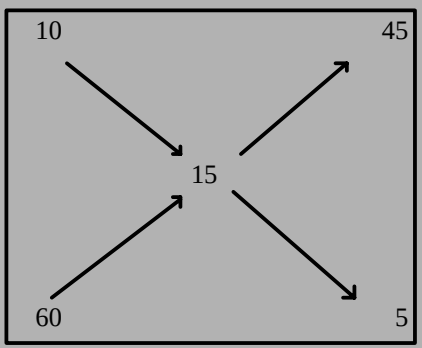
Many types of liquid food are transported through pipes during processing, and powders and small-particulate foods are more easily handled as fluids (by fluidisation). Gases obey the same laws as liquids and, for the purposes of calculations, gases are treated as compressible fluids. Properties of selected fluids are shown in Table 1.2. The study of fluids is therefore of great importance in food processing. It is divided into fluid statics (stationary fluids) and fluid dynamics (moving fluids).

A property of static liquids is the pressure that they exert on the containing vessel. The pressure is related to the density of the liquid and the depth or the mass of liquid in the vessel. Liquids at the base of a vessel are at a higher pressure than at the surface, owing to the weight of liquid above (the *hydrostatic head*). This is important in the design of holding tanks and processing vessel, to ensure that the vessel is constructed using materials of

Sample problem 1.3
Use a Pearson Square to calculate the amounts of orange juice (10% sugar content) and sugar syrup (60% sugar content) needed to produce 50 kg of fruit squash containing 15% sugar.

Solution to Sample problem 1.3

Orange juice



Sugar syrup

The result indicates that 45 kg of orange juice should be mixed with 5 kg of sugar syrup.

adequate strength. A large hydrostatic head also affects the boiling point of liquids, which is important in the design of some types of evaporation equipment (Chapter 13).

When a fluid flows through pipes or processing equipment (Fig. 1.6), there is a loss of energy and a drop in pressure which are due to frictional resistance to flow. These friction losses and changes in the potential energy, kinetic energy and pressure energy are described in detail in the food engineering texts referenced at the start of this chapter and by Loncin and Merson (1979). The loss of pressure in pipes is determined by a number of factors including the density and viscosity of the fluid, the length and diameter of the pipe and the number of bends, valves, etc., in the pipeline. To overcome this loss in energy, it is necessary to apply power using pumps to transport the fluid. The amount of power required is determined by the viscosity of the fluid (Section 1.1.2), the size of the pipework, the number of bends and fittings, and the height and distance that the fluid is to be moved.

To calculate the energy balance when a liquid flows through pipework, the effect of valves or bends on the flow rate, or the pressure developed by a pump, use is made of *Bernoulli's equation*, which is a statement of the conservation of energy.

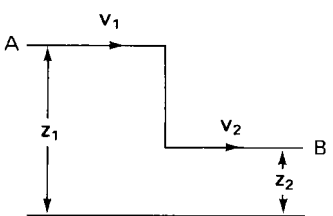


Fig 1.6 Application of Bernoulli's equation to frictionless fluid flow.

$$\frac{P_1}{\rho_1} + \frac{v_1^2}{2} = z_1 g = \frac{P_2}{\rho_2} + \frac{v_2^2}{2} + z_2 g \qquad 1.8$$

where P (Pa) = the pressure, ρ (kg m^{-3}) = the fluid density, g (= 9.81 m s^{-1}) = acceleration due to gravity, v (m s^{-1}) = the velocity of the fluid and z (m) = the height. The subscript 1 indicates the first position in the pipework and the subscript 2 the second position in the pipework.

In any system in which fluids flow, there exists a *boundary film* (or surface film) of

Sample problem 1.4

A 20% sucrose solution flows from a mixing tank at 50 kPa through an horizontal pipe 5 cm in diameter at 25 m^3 h^{-1}. If the pipe diameter reduces to 3 cm, calculate the new pressure in the pipe. (The density of sucrose solution is 1070 kg m^{-3} (Table 1.2).)

Solution to Sample problem 1.4

$$\text{Flow rate} = \frac{25}{3600}\text{m}^3\,\text{s}^{-1}$$

$$= 6.94 \times 10^{-3}\,\text{m}^3\,\text{s}^{-1}$$

$$\text{Area of pipe 5 cm in diameter} = \frac{\pi}{4}D^2$$

$$= \frac{3.142}{4}(0.05)^2$$

$$= 1.96 \times 10^{-3}\text{m}^2$$

$$\text{Velocity of Flow} = \frac{6.94 \times 10^{-3}}{1.96 \times 10^{-3}}$$

$$= 3.54\text{m s}^{-1}$$

Area of pipe 3 cm in diameter $= 7.07 \times 10^{-4}$ m^2

$$\text{Velocity of Flow} = \frac{6.94 \times 10^{-3}}{7.70 \times 10^{-4}}$$

$$= 9.81\text{m s}^{-1}$$

Using equation (1.8),

$$\frac{P_1}{\rho_1} + \frac{v_1^2}{2} + z_1 g = \frac{P_2}{\rho_2} + \frac{v_2^2}{2} + z_2 g$$

$$\frac{3.54^2}{2} + \frac{50 \times 10^3}{1070} + 0 = \frac{P_2}{1070} + \frac{9.81^2}{2} + 0$$

Therefore,

$$P_2 = 56\,656\text{ Pa}$$

$$= 56.7\text{ kPa}$$

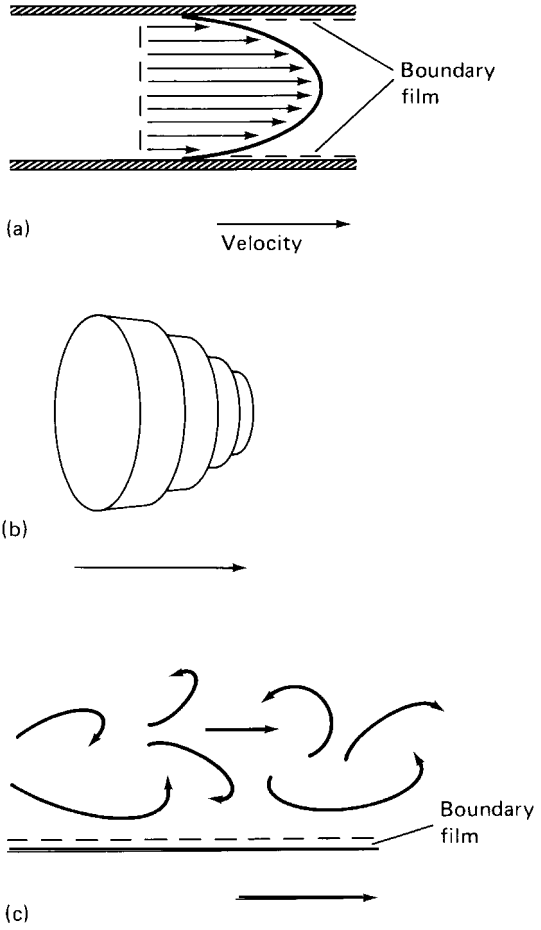


Fig. 1.7 Fluid flow: (a) velocity distribution and boundary layer; (b) streamline flow; (c) turbulent flow.

fluid next to the surface over which the fluid flows (Fig. 1.7(a)). The thickness of the boundary film is influenced by a number of factors, including the velocity, viscosity, density and temperature of the fluid. Fluids which have a low flow rate or high viscosity may be thought of as a series of layers which move over one another without mixing (Fig. 1.7(b)). This produces movement of the fluid, which is termed *streamline* (or laminar) flow. In a pipe, the velocity of the fluid is highest at the centre and zero at the pipe wall. Above a certain flow rate, which is determined by the nature of fluid and the pipe, the layers of liquid mix together and *turbulent* flow is established (Fig. 1.7 (c)) in the bulk of the fluid, although the flow remains streamline in the boundary film. Higher flowrates produce more turbulent flow and hence thinner boundary films.

Fluid flow is characterised by a dimensionless group (Appendix D) named the *Reynolds number* (Re). This is calculated using

$$\mathrm{Re} = \frac{Dv\rho}{\mu} \qquad 1.9$$

where D (m) = the diameter of the pipe, v (m s^{-1}) = average velocity, ρ (kg m^{-3}) = fluid density and μ (N s m^{-2}) = fluid viscosity.

A Reynolds number of less than 2100 describes streamline flow and a Reynolds number of more than 4000 describes turbulent flow. For Reynolds numbers between 2100 and 4000, *transitional* flow is present, which can be either laminar or turbulent at different times. These different flow characteristics have important implications for heat transfer and mixing operations; turbulent flow produces thinner boundary layers, which in turn permit higher rates of heat transfer. The implications of this for the design and performance of equipment are discussed in Chapters 11–13 and 17 (for liquids through pipes or over metal plates) and Chapters 15, 16 and 21 (for air moving over the surface of food or metal). The Reynolds number can also be used to determine the power requirements for pumps and mixers used for blending and mixing operations (Chapter 5).

In turbulent flow, particles of fluid move in all directions and solids are retained in suspension more readily. This reduces the formation of deposits on heat exchangers and prevents solids from settling out in pipework. Streamline flow produces a larger range of residence times for individual particles flowing in a tube. This is especially important when calculating the residence time required for heat treatment of liquid foods, as it is necessary to ensure that all parts of the food receive the required amount of heat. This aspect is discussed in more detail in relation to ohmic heating (Chapter 18). Turbulent flow causes higher friction losses than streamline flow does and therefore requires higher energy inputs from pumps.

Sample problem 1.5

Two fluids, milk and rape-seed oil, are flowing along pipes of the same diameter (5 cm) at 20°C and at the same flow velocity of $3\ \mathrm{m\ s^{-1}}$. Determine whether the flow is streamline or turbulent in each fluid. (Physical properties of milk and rape-seed oil are shown in Table 1.2.)

Solution to Sample problem 1.5

For milk from Table 1.2, $\mu = 2.10 \times 10^{-3}\ \mathrm{N\ s\ m^{-2}}$ and $\rho = 1030\ \mathrm{kg\ m^{-3}}$. From equation (1.9),

$$\mathrm{Re} = \frac{Dv\rho}{\mu}$$

Therefore,

$$\mathrm{Re} = \frac{0.05 \times 3 \times 1030}{2.1 \times 10^{-3}}$$

$$= 73\,571$$

Thus the flow is turbulent (because Re is greater than 4000).

For rape-seed oil, from Table 1.2, $\mu = 118 \times 10^{-3}\mathrm{N\ s\ m^{-2}}$ and $\rho = 900\ \mathrm{kg\ m^{-3}}$. Therefore

$$\mathrm{Re} = \frac{0.05 \times 3 \times 900}{118 \times 10^{-3}}$$

$$= 1144$$

Thus the flow is streamline (because Re is less than 2100).

1.3.1 Fluid flow through fluidised beds

When air passes upwards through a bed of food, the particles create a resistance to the flow of air and reduce the area available for it to flow through the bed. As the air velocity is increased, a point is reached where the weight of the food is just balanced by the force of the air, and the bed becomes fluidised (for example fluidised-bed drying (Chapter 15)). If the velocity is increased further, the bed becomes more open (the *voidage* is increased), until eventually the particles are conveyed in the fluid stream (for example pneumatic separation (Chapter 3), pneumatic drying (Chapter 15), or pneumatic conveying (Chapter 26)). The velocity of the air needed to achieve fluidisation of spherical particles is calculated using

$$v_f = \frac{(\rho_s - \rho)g}{\mu} \frac{d^2\epsilon^3}{180(1-\epsilon)} \qquad 1.10$$

where v_f (m s^{-1}) = fluidisation velocity, ρ_s (kg m^{-3}) = density of the solid particles, ρ (kg m^{-3}) = density of the fluid, g (m s^{-2}) = acceleration due to gravity, μ (N s m^{-2}) = viscosity of the fluid, d (m) = diameter of the particles, ϵ = the voidage of the bed.

Formulae for foods of other shapes are described by Kunii and Levenspiel (1969). The minimum air velocity needed to convey particles is found using

$$v_e = \sqrt{\left[\frac{4d(\rho_s - \rho)}{3C_d\rho}\right]} \qquad 1.11$$

where v_e (m s^{-1}) = minimum air velocity and C_d (= 0.44 for Re = 500–200 000) = the drag coefficient.

Sample problem 1.6

Peas which have an average diameter of 6 mm and a density of 880 kg m^{-3} are dried in a fluidised-bed drier (Chapter 15). The minimum voidage is 0.4 and the cross-sectional area of the bed is 0.25 m^2. Calculate the minimum air velocity needed to fluidise the bed if the air density is 0.96 kg m^{-3} and the air viscosity is 2.15×10^{-5} N s m^{-2}.

Solution to Sample problem 1.6

From equation (1.10),

$$v_f = \frac{(880 - 0.96)9.81}{2.15 \times 10^{-5}} \frac{(0.006)^2(0.4)^3}{180(1-0.4)}$$
$$= 8.5 \text{ m s}^{-1}$$

1.4 Heat transfer

Most unit operations in food processing involve the transfer of heat into or out of a food. There are three ways in which heat may be transferred: by radiation, by conduction and by convection. *Radiation*, which is described in detail in Chapter 18, is the transfer of heat by electromagnetic waves (see Chapter 18, Fig. 18.1), for example in an electric grill. *Conduction* is the movement of heat by direct transfer of molecular energy within solids (for example through metal containers or solid foods). *Convection* is the transfer of heat by groups of molecules that move as a result of differences in density (for example in heated

air) or as a result of agitation (for example in stirred liquids). In the majority of applications all three types of heat transfer occur simultaneously but one type may be more important than others in particular applications. Further details are given by Hayhurst (1997).

1.4.1 Energy balances

An energy balance states that 'the amount of heat or mechanical energy entering a process = the total energy leaving with the products and wastes + stored energy + energy lost to the surroundings'. If heat losses are minimised, energy losses to the surroundings may be ignored for approximate solutions to calculation of, for example, the quantity of steam, hot air or refrigerant required. For more accurate solutions, compensation should be made for heat losses.

1.4.2 Mechanisms of heat transfer

Steady-state heat transfer takes place when there is a constant temperature difference between two materials. The amount of heat entering a material equals the amount of heat leaving, and there is no change in temperature of the material. This occurs for example when heat is transferred through the wall of a cold store if the store temperature and ambient temperature are constant (Chapter 19), and in continuous processes once operating conditions have stabilised. However, in the majority of food-processing applications the temperature of the food and/or the heating or cooling medium are constantly changing, and *unsteady-state* heat transfer is more commonly found. Calculations of heat transfer under these conditions are extremely complicated but are simplified by making a number of assumptions and using prepared charts and computer models to give approximate solutions (Toledo, 1999).

Steady-state conduction

The rate at which heat is transferred by conduction is determined by the temperature difference between the food and the heating or cooling medium, and the total resistance to heat transfer. The resistance to heat transfer is expressed as the conductance of a material, or more usefully as the reciprocal which is termed the *thermal conductivity*. Thermal conductivities of some materials found in food processing are shown in Tables 1.2 and 1.5.

Under steady-state conditions the rate of heat transfer is calculated using

$$Q = \frac{kA(\theta_1 - \theta_2)}{x} \qquad 1.12$$

where Q ($\mathrm{J\,s^{-1}}$) = rate of heat transfer, k ($\mathrm{J\,m^{-1}\,s^{-1}\,K^{-1}}$ or $\mathrm{W\,m^{-1}\,K^{-1}}$) = thermal conductivity, A ($\mathrm{m^2}$) = surface area, $\theta_1 - \theta_2$ (°C or K) = temperature difference and x (m) = thickness of the material. $(\theta_1 - \theta_2)/x$ is also known as the temperature gradient.

Although, for example, stainless steel conducts heat ten times less well than aluminium (Table 1.5), the difference is small compared to the low thermal conductivity of foods (twenty to thirty times smaller than steel) and does not limit the rate of heat transfer. Stainless steel is much less reactive than other metals, particularly with acidic foods, and is therefore used in most food-processing equipment that comes into contact with foods.

The thermal conductivity of foods is influenced by a number of factors concerned with the nature of the food (for example cell structure, the amount of air trapped between the

Table 1.5 Thermal conductivity of selected foods and other materials

Type of material	Thermal conductivity ($W\ m^{-1}\ K^{-1}$)	Temperature of measurement (°C)
Construction materials		
Aluminium	220	0
Copper	388	0
Stainless steel	21	20
Other metals	45–400	0
Brick	0.69	20
Concrete	0.87	20
Foods		
Olive oil[a]	0.17	20
Whole milk[a]	0.56	20
Freeze-dried foods	0.01–0.04	0
Frozen beef[b]	1.30	−10
Pork (lean)[b]	0.48	3.8
Frozen cod	1.66	−10
Apple juice	0.56	20
Orange	0.41	0–15
Green beans	0.80	−12.1
Cauliflower	0.80	−6.6
Egg	0.96	−8
Ice	2.25	0
Water[a]	0.57	0
Packaging materials		
Cardboard	0.07	20
Glass, soda	0.52	20
Polyethylene	0.55	20
Poly(vinyl chloride)	0.29	20
Insulating materials		
Polystyrene foam	0.036	0
Polyurethane foam	0.026	0
Other types	0.026–0.052	30

[a] Assuming convection currents are absent.
[b] Heat flow parallel to fibres.
From Earle (1983), Lewis (1987) and Woodams and Nowrey (1968).

cells, and the moisture content), and the temperature and pressure of the surroundings. A reduction in moisture content causes a substantial reduction in thermal conductivity. This has important implications in unit operations which involve conduction of heat through food to remove water (for example drying (Chapter 15), frying (Chapter 17) and freeze drying (Chapter 22)). In freeze drying the reduction in atmospheric pressure also influences the thermal conductivity of the food. Ice has a higher thermal conductivity than water and this is important in determining the rate of freezing and thawing (Chapter 21). (A sample problem indicating the importance of thermal conductivity is shown in Chapter 17.)

Unsteady-state conduction

During processing, the temperature at a given point within a food depends on the rate of heating or cooling and the position in the food. The temperature therefore changes continuously. The factors that influence the temperature change are:

- the temperature of the heating medium

Table 1.6 Specific heat of selected foods and other materials

Food/material	Specific heat (kJ kg^{-1} K^{-1})	Temperature (°C)
Aluminium	0.89	20
Apples	3.59	ambient
Apples	1.88	frozen
Cod	3.76	ambient
Cod	2.05	frozen
Copper	0.38	20
Ice	2.04	0
Lamb	2.80	ambient
Lamb	1.25	frozen
Potatoes	3.43	ambient
Potatoes	1.80	frozen
Stainless steel	0.46	20
Water	4.18	15
Water vapour	2.05	100

Adapted from data of Peleg and Bagley (1983), Jowitt *et al.* (1983) and Polley *et al.* (1980).

- the thermal conductivity of the food
- the specific heat of the food.

Examples of the specific heat of foods are given in Tables 1.2 and 1.6. Thermal diffusivity is related to the thermal conductivity, specific heat and density of a food by

$$a = \frac{k}{\rho c} \qquad 1.13$$

where a ($m_2\,s^{-1}$) = the thermal diffusivity, ρ ($kg\,m^{-3}$) = density, c ($J\,kg^{-1}\,K^{-1}$) = specific heat capacity and k ($W\,m^{-1}\,K^{-1}$) = thermal conductivity.

The basic equation for unsteady-state heat transfer in a single direction (x) is

$$\frac{d\theta}{dt} = \frac{k}{\rho c}\frac{d^2\theta}{dx^2} \qquad 1.14$$

where $d\theta/dt$ = change in temperature with time.

Examples of solutions to this equation for simple shapes (for example a slab, cylinder or sphere) are described by Earle (1983), Jackson and Lamb (1981) and Loncin and Merson (1979).

Convection

When a fluid changes temperature, the resulting changes in density establish natural-convection currents. Examples include natural-circulation evaporators (Chapter 13), air movement in chest freezers (Chapter 21), and movement of liquids inside cans during sterilisation (Chapter 12). Forced convection takes place when a stirrer or fan is used to agitate the fluid. This reduces the boundary film thickness to produce higher rates of heat transfer and a more rapid temperature redistribution. Consequently, forced convection is more commonly used than natural convection in food processing. Examples of forced convection include mixers (Chapter 5), fluidised-bed driers (Chapter 15), air blast freezers (Chapter 21) and liquids pumped through heat exchangers (Chapters 11–13).

When liquids or gases are used as heating or cooling media, the rate of heat transfer from the fluid to the surface of a food is found using

$$Q = h_s A(\theta_b - \theta_s) \quad 1.15$$

where Q (J s^{-1}) = rate of heat transfer, A (m^2) = surface area, θ_s (K) = surface temperature, θ_b (K) = bulk fluid temperature and h_s (W m^{-2} K^{-1}) = surface (or film) heat transfer coefficient.

The *surface heat transfer coefficient* is a measure of the resistance to heat flow, caused by the boundary film, and is therefore equivalent to the term k/x in the conduction equation (equation 1.12) (Polley *et al.*, 1980). It is higher in turbulent flow than in streamline flow. Typical values of h_s are given in Table 1.7.

These data indicate that heat transfer through air is lower than through liquids and higher rates of heat transfer are obtained by moving air than still air. Larger heat exchangers are therefore necessary when air is used for heating (Chapters 15 and 16) or cooling (Chapters 19 and 21) compared to those needed for liquids. Condensing steam produces higher rates of heat transfer than hot water at the same temperature and the presence of air in steam reduces the rate of heat transfer. This has important implications for canning (Chapter 12) as any air in the steam lowers the temperature and hence lowers that amount of heat received by the food. Both thermometers and pressure gauges are therefore needed to assess whether steam is saturated.

The surface heat transfer coefficient is related to the physical properties of a fluid (for example density, viscosity, specific heat), gravity (which causes circulation due to changes in density), temperature difference and the length or diameter of the container under investigation. The formulae which relate these factors are expressed as dimensionless numbers (Appendix D) as follows:

$$\text{Nusselt number Nu} = \frac{h_c D}{k} \quad 1.16$$

$$\text{Prandtl number Pr} = \frac{c_p \mu}{k} \quad 1.17$$

$$\text{Grashof number Gr} = \frac{D^3 \rho^2 g \beta \Delta\theta}{\mu^2} \quad 1.18$$

where h_c (W m^{-2} K^{-1}) = convection heat transfer coefficient at the solid-liquid interface, D (m) = the characteristic dimension (length or diameter), k (W m^{-1} K^{-1}) = thermal con-

Table 1.7 Values of surface heat transfer coefficients

	Surface heat transfer coefficient h_s (W m^{-2} K^{-1})	Typical applications
Boiling liquids	2400–60 000	Evaporation
Condensing saturated steam	12 000	Canning, evaporation
Condensing steam		
With 3% air	3500	Canning
With 6% air	1200	
Condensing ammonia	6000	Freezing, chilling
Liquid flowing through pipes		
low viscosity	1200–6000	Pasteurisation
high viscosity	120–1200	Evaporation
Moving air (3 m s^{-1})	30	Freezing, baking
Still air	6	Cold stores

Adapted from Loncin and Merson (1979) and Earle (1983).

ductivity of the fluid, c_p (J kg^{-1} K^{-1}) = specific heat at constant pressure, ρ (kg m^{-3}) = density, μ (N s m^{-2}) = viscosity, g (m s^{-2}) = acceleration due to gravity, β (m m^{-1} K^{-1}) = coefficient of thermal expansion, $\Delta\theta$ (K) = temperature difference and v (m s^{-1}) = velocity.

For streamline flow through pipes,

$$\mathrm{Nu} = 1.62(\mathrm{Re}\ \mathrm{Pr}\frac{D}{L})^{0.33} \qquad 1.19$$

where L (m) = length of pipe, when Re Pr $D/L > 120$ and all physical properties are measured at the mean bulk temperature of the fluid.

For turbulent flow through pipes,

$$\mathrm{Nu} = 0.023(\mathrm{Re})^{0.8}(\mathrm{Pr})^{n} \qquad 1.20$$

where $n = 0.4$ for heating or $n = 0.3$ for cooling, when Re $>$ 10 000, viscosity is measured at the mean *film* temperature and other physical properties are measured at the mean bulk temperature of the fluid.

The Grashof number is used for natural convection when there is no turbulence in the fluid. Formulae for other types of flow conditions and different vessels are described by Loncin and Merson (1979), Jackson and Lamb (1981) and Earle (1983).

Sample problem 1.7

Whole milk is cooled in the pipes of a tubular heat exchanger from 30ºC to 10ºC, by water at 1ºC. The pipe diameter is 5 cm and the milk flow velocity is 1.0 m s^{-1}. Calculate the heat transfer coefficient for the milk using physical properties described in Table 1.2.

Solution to Sample problem 1.7

$$\text{Mean bulk temperature} = \frac{30 + 10}{2}$$
$$= 20^{\circ}\mathrm{C}$$

From Table 1.2, $\rho = 1030$ kg m^{-3}, $c = 3.9$ kJ kg^{-1}ºK^{-1} and

$$k = 0.56\ \mathrm{W\ m^{-2}\ ^{\circ}K^{-1}}.$$

If Re $>$ 10 000 viscosity should be measured at the mean temperature.

$$\text{Mean film temperature} = \frac{1 + \frac{1}{2}(30 + 10)}{2}$$
$$= 10.5^{\circ}\mathrm{C}$$

For milk at 10.5ºC, $\mu = 2.8 \times 10^{-3}$N s m^{-2}. From equation (1.9),

$$\mathrm{Re} = \frac{Dv\rho}{\mu}$$
$$= \frac{0.05 \times 1.0 \times 1030}{2.8 \times 10^{-3}}$$
$$= 18\,393$$

From equation (1.17)

$$\mathrm{Pr} = \frac{c_p \mu}{k}$$

$$= \frac{(3.9 \times 10^3)(2.8 \times 10^{-3})}{0.56}$$

$$= 19.5$$

From equations (1.16) and (1.20)

$$\mathrm{Nu} = \frac{h_c D}{k}$$

$$= 0.023(\mathrm{Re})^{0.8}(\mathrm{Pr})^{0.33}$$

Therefore,

$$h_c = 0.023 \frac{k}{D}(\mathrm{Re})^{0.8}(\mathrm{Pr})^{0.33}.$$

$$= 0.023 \frac{0.56}{0.05}(18\,393)^{0.8}(19.5)^{0.33}$$

$$= 1768 \text{ W m}^{-2\,\circ}\text{K}^{-1}$$

Most cases of heat transfer in food processing involve heat transfer through a number of different materials. For example heat transfer in a heat exchanger from a hot fluid, through the wall of a vessel to a second fluid is shown in Fig. 1.8.

The overall temperature difference is found using

$$\theta_a - \theta_b = \frac{Q}{A}\left(\frac{1}{h_a}\frac{x}{k} + \frac{1}{h_b}\right) \qquad 1.21$$

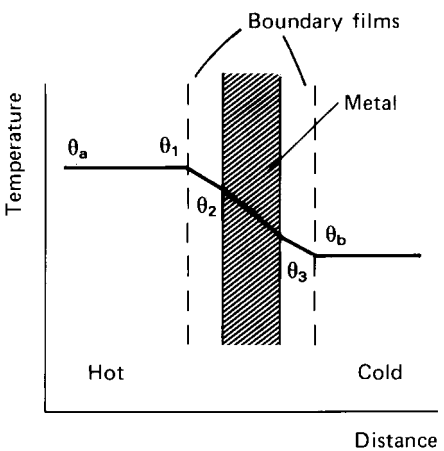


Fig. 1.8 Temperature changes from a hot liquid through a vessel wall to a cold liquid.

Table 1.8 OHTCs in food processing

Heat transfer fluids	Example	OHTC (W m^{-2} K^{-1})
Hot water–air	Air heater	10–50
Viscous liquid–hot water	Jacketed vessel	100
Viscous liquid–hot water	Agitated jacketed vessel	500
Viscous liquid–steam	Evaporator	500
Non-viscous liquid–steam	Evaporator	1000–3000
Flue gas–water	Boiler	5–50
Evaporating ammonia–water	Chilled water plant	500

Adapted from Lewis (1990).

The unknown wall temperatures θ_2 and θ_3 are therefore not required and all factors to solve the equation can be measured.

The sum of the resistances to heat flow is termed the *overall heat transfer coefficient* (OHTC) U and the rate of heat transfer may then be expressed as

$$Q = UA(\theta_a - \theta_b) \qquad 1.22$$

The OHTC is an important term which is used, for example, to indicate the effectiveness of heating or cooling in different types of processing equipment. Examples are shown in Table 1.8.

Liquids can be made to flow in either the same direction or in opposite directions in a heat exchanger. Counter-current flow of fluids (Fig. 1.9) has a higher heat transfer efficiency than co-current (or 'parallel') flow and is therefore widely used in heat exchangers (for example Chapters 10–21). However, the temperature difference varies at different points in the heat exchanger and it is necessary to use a logarithmic mean temperature difference in calculations (equation 1.23):

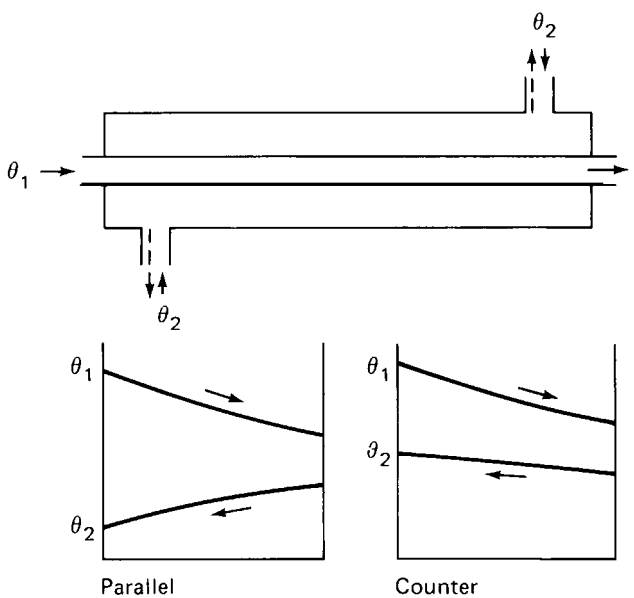


Fig. 1.9 Parallel and counter-current flow through a heat exchanger.

$$\Delta\theta_m = \frac{\Delta\theta_1 - \Delta\theta_2}{\ln(\Delta\theta_1/\theta_2)} \quad 1.23$$

where θ_1 is higher than θ_2.

Related sample problems are shown in Chapters 11, 13 and 15.

The heating time in batch processing is found using

$$t = \frac{mc}{UA}\ln\left(\frac{\theta_h - \theta_i}{\theta_h - \theta_f}\right) \quad 1.24$$

where m (kg) = the mass, c (J kg^{-1} ^{o}K^{-1}) = specific heat capacity, θ_h (^{o}C) = temperature of the heating medium, θ_i (^{o}C) = initial temperature, θ_f (^{o}C) = final temperature, A (m^2) = surface area and U (W m^{-2} K^{-1}) = OHTC.

Sample problem 1.8

In the counter-current heat exchanger shown in Fig. 1.9, milk is cooled from 73ºC to 38ºC at the rate of 2500 kg h^{-1}, using water at 15ºC which leaves the heat exchanger at 40ºC. The pipework 2.5 cm in diameter is constructed from stainless steel 3 mm thick; the surface film heat transfer coefficients are 1200 W m^{-2} K^{-1} on the milk side and 3000 W m^{-2} K^{-1} on the water side of the pipe. Calculate the OHTC and the length of pipe required.

Solution to Sample problem 1.8

To find the OHTC, from equations (1.21) and (1.22),

$$\begin{aligned}\frac{1}{U} &= \frac{1}{h_a} + \frac{x}{k} + \frac{1}{h_b}\\ &= \frac{1}{1200} + \frac{3\times10^{-3}}{21} + \frac{1}{3000}\\ &= 1.3\times10^{-3}\end{aligned}$$

Therefore the OHTC is

$$U = 769.2\ \text{W m}^{-2}\,^{\circ}\text{K}^{-1}$$

To find the length of pipe required, we proceed as follows.

From equations (1.22) and (1.23),

$$Q = UA\Delta\theta_m$$

and

$$\begin{aligned}\Delta\theta_m &= \frac{\Delta\theta_1 - \Delta\theta_2}{\ln\,(\Delta\theta_1/\Delta\theta_2)}\\ &= \frac{(73-40)/(38-15)}{\ln[(73-40)/(38-15)]}\\ &= 27.8^{\circ}\text{C}\end{aligned}$$

Now Q is the heat removed from the milk which equals $mc_p(\theta_a - \theta_b)$. From Table 1.2, c_p = 3.9 kJ kg^{-1} ^{o}K^{-1}. Therefore,

$$Q = \frac{2500}{3600}(3.9 \times 10^3)(73 - 38)$$

$$= 9.48 \times 10^4 \text{J}$$

Now the area is

$$A = \frac{Q}{U\Delta\theta_m}$$

$$= \frac{9.48 \times 10^4}{769.2 \times 27.8}$$

$$= 4.4 \text{ m}^2$$

Also

$$A = \pi DL$$

Therefore the length of pipe is

$$L = \frac{A}{\pi D}$$

$$= \frac{4.4}{3.142 \times 0.025} = 56 \text{ m}$$

Unsteady-state heat transfer by conduction and convection

When a solid piece of food is heated or cooled by a fluid, the resistances to heat transfer are the surface heat transfer coefficient and the thermal conductivity of the food. These two factors are related by the *Biot Number* Bi:

$$\text{Bi} = \frac{h\delta}{k} \qquad 1.25$$

where h (W m^{-2} K^{-1}) = heat transfer coefficient, δ = the characteristic 'half dimension' (e.g. radius of a sphere or cylinder, half thickness of a slab) and k (W m^{-1} K^{-1}) = thermal conductivity.

At small Bi values (less than 0.2) the surface film is the main resistance to heat flow and the time required to heat the solid food is found using equation (1.24), using the film heat transfer coefficient h_s instead of U. However, in most applications the thermal conductivity of the food limits the rate of heat transfer (Bi > 0.2). These calculations are complex, and a series of charts is available to solve the unsteady-state equations for simple shaped foods (Fig. 1.10). The charts relate the Biot number (Bi) (equation 1.25), the *temperature factor* (the fraction of the temperature change that remains to be accomplished (equation (1.26))) and the *Fourier number* Fo (a dimensionless number which relates the thermal diffusivity, the size of the piece and the time of heating or cooling (equation (1.27))).

$$\frac{\theta_h - \theta_f}{\theta_h - \theta_i} \qquad 1.26$$

where the subscript h indicates the heating medium, the subscript f the final value and the subscript i the initial value.

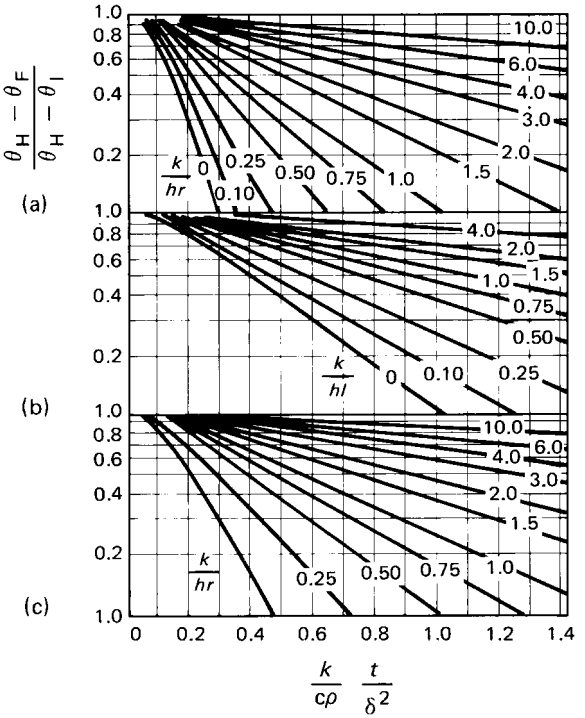


Fig. 1.10 Chart for unsteady-state heat transfer: (a) sphere; (b) slab; (c) cylinder. (After Henderson and Perry (1955).)

$$\text{Fo} = \frac{k}{c\rho}\frac{t}{\delta^2} \qquad 1.27$$

Examples of unsteady-state calculations are shown below and in Chapter 19. More complex calculations are described by Lewis (1990), Earle (1983) and Jackson and Lamb (1981).

Sample problem 1.9
Peas which have an average diameter of 6 mm are blanched to give a temperature of 85ºC at the centre. The initial temperature of the peas is 15ºC and the temperature of the blancher water is 95ºC. Calculate the time required, assuming that the heat transfer coefficient is $1200\,\text{W m}^{-2}\,\text{K}^{-1}$ and, for peas, the thermal conductivity is $0.35\,\text{W m}^{-1}\,\text{K}^{-1}$, the specific heat is $3.3\,\text{KJ kg}^{-1}\,\text{K}^{-1}$, and the density is $980\,\text{kg m}^{-3}$.

Solution to Sample problem 1.9
From equation (1.25),

$$\text{Bi} = \frac{h\delta}{k}$$

$$= \frac{1200(3 \times 10^{-3})}{0.35}$$

$$= 10.3$$

Therefore,

$$\frac{k}{h\delta} = 0.097$$

From equation (1.26),

$$\frac{\theta_h - \theta_f}{\theta_h - \theta_i} = \frac{95 - 85}{95 - 15}$$

$$= 0.125$$

From the chart for a sphere (Fig. 1.10), Fo = 0.32. From equation (1.27),

$$\text{Fo} = \frac{k}{c\rho}\frac{t}{\delta^2}$$

$$= 0.32$$

Therefore,

$$\text{time } t = 0.32\frac{c\rho\delta^2}{k}$$

$$= \frac{0.32(3.3 \times 10^3)980(3 \times 10^{-3})^2}{0.35}$$

$$= 26.6 \text{ s}$$

1.4.3 Sources of heat and methods of application to foods

The cost of energy for heating has become one of the major considerations in the selection of processing methods and ultimately in the cost of the processed food and the profitability of the operation. Different fuels have specific advantages and limitations in terms of cost, safety, risk of contamination of the food, flexibility of use, and capital and operating costs for heat transfer equipment. The following sources of energy are used in food processing:

- electricity
- gas (natural and liquid petroleum gas)
- liquid fuel oil.

Solid fuels (anthracite, coal, wood and charcoal) are only used to a small extent. The advantages and limitations of each type of energy source are shown in Table 1.9.

Direct heating methods

In direct methods the heat and products of combustion from the burning fuel come directly into contact with the food. There is an obvious risk of contamination of the food by odours or incompletely burned fuel and, for this reason, only gas and, to a lesser extent, liquid fuels are used. Applications include kiln driers (Chapter 15) and baking ovens (Chapter 16). These direct methods should not be confused with 'direct' steam injection where the steam is produced in a separate location from the processing plant. Electricity is not a fuel in the same sense as the other types described above. It is generated by steam turbines heated by a primary fuel (for example coal or fuel oil) or by hydro-power or nuclear fission. However, electrical energy is also said to be used directly in dielectric heating or microwave heating (Chapter 18).

Table 1.9 Advantages and limitations of different energy sources for food processing

	Electricity	Gas	Liquid fuel	Solid fuel
Energy per unit mass or volume	Not applicable	Low[a]	High[b]	Moderate to high[c]
Cost per kilojoule of energy	High	Low	Low	Low
Heat transfer equipment cost	Low	Low	High	High
Efficiency of heating[d]	High	Moderate to high	Moderate to low	Low
Flexibility of use	High	High	Low	Low
Fire or explosion hazard	Low	High	Low	Low
Risk of food contamination	Low	Low	High	High
Labour and handling costs	Low	Low	Low	High

[a] Heating value for gas is $1.17 \times 10^3 - 4.78 \times 10^3$ kJ kg^{-1}.
[b] Heating value for oil is $8.6 \times 10^3 - 9.3 \times 10^3$ kJ kg^{-1}.
[c] Heating value for coal is $5.26 \times 10^3 - 6.7 \times 10^3$ kJ kg^{-1} and for wood is $3.8 \times 10^3 - 5.26 \times 10^3$ kJ kg^{-1}.
[d] Efficiency defined as the amount of energy used to heat the product divided by the amount of energy supplied.
Data compiled from Farrall (1979) and Whitman *et al*. (1981).

Indirect methods

Indirect cooling of foods is described in Chapters 19 and 21. Indirect heating methods employ a heat exchanger to separate the food from the products of combustion. At its simplest an indirect system consists of burning fuel beneath a metal plate and heating by energy radiated from the plate. The most common type of indirect-heating system used in food processing is steam or hot water generated by a heat exchanger (a boiler) located away from the processing area. A second heat exchanger transfers the heat from the steam to the food under controlled conditions or the steam is injected into the food. A variation on this system involves an additional heat exchanger which transfers heat from the steam to air in order to dry foods or to heat them under dry conditions. The advantages of steam for process heating, and a description of equipment for producing and handling different types of steam, are given by Farrall (1979). Properties of steam are discussed by Brennan *et al*. (1990) and Toledo (1999) and the temperatures of saturated steam are shown in Chapter 12 (Table 12.6).

Indirect electrical heating uses resistance heaters or infrared heaters. Resistance heaters are nickel-chromium wires contained in solid plates or coils, which are attached to the walls of vessels, in flexible jackets which wrap around vessels, or in immersion heaters which are submerged in the food. These types of heater are used for localised or intermittent heating. Infrared heaters are described in Chapter 18.

1.4.4 Energy conservation

A survey of food industries within the European Union found large variations in energy consumption in different types of processing; flour milling has the lowest energy input (586 MJ per tonne of product) and cocoa and chocolate processing the highest (8335 MJ per tonne of product) (Whitman *et al.*, 1981). A comparison of energy inputs in different types of potato processing is shown in Table 1.10.

In all types of food processing, most of the energy (58%; range, 40–80%) is used for actual processing, but significant amounts are also used for packaging (11%; range, 15–40%), transport (12%; range, 0.56–30%), cleaning water (15%) and storage (up to 85% of total energy input for deep-frozen foods). Boiler fuel accounts for 8.6–97.2% of the total

Table 1.10 Comparative energy inputs in potato processing

	Energy input (MJ t^{-1})				
	Hot air dried	Freeze dried	Canned	Retort pouched	Frozen
Processing	8	42	6	4	5
Packaging	1	2	14	7	7
Storage	0.1	0.1	0.1	0.1	32
(Home)	0	0	0	0	(24)
Transport	2	2	8	8	8
Total	11	46	28	19	52

Reported in Kelford (1982).

energy input to a factory, but boiler utilisation averages only 45%, owing to the practice of keeping the boiler on low heat during periods of low production rates (for example evening and night shifts). In addition an average of only half of the condensate is returned to the boiler and the heat contained in the condensate is therefore wasted. On average 40% of the heat lost from factories is contained in vapours and steam and a further 10–20% is lost in hot water (Whitman *et al.*, 1981).

It is clear from this and other data (for example Kefford, 1982), that potentially the main energy savings in food processing are associated with boiler operation, the supply of steam or hot air, and re-use of waste heat. An *energy audit* is used to identify specific areas and equipment within a factory where energy savings can be made (Boardman, 1986). In summary, boiler operation is improved by:

- returning condensate as feed water
- pre-heating air for combustion of fuel
- insulating the boiler
- recovering heat from the flue.

Computer control of boiler operation is described by Wells and Swientek (1983). Energy savings in steam supply to processing areas are achieved by:

- proper insulation of steam and hot-water pipes
- minimising steam leaks
- fitting steam traps.

Individual processing equipment is designed for energy saving. Examples include regeneration of heat in heat exchangers (Delashmit *et al.*, 1983) (examples in Chapters 10, 11, 16 and 17), multiple-effect or vapour recompression systems (Chapter 13) and automatic defrosting and correct insulation of freezing equipment (Chapters 19–21). Microprocessor control of processing equipment (Chapter 2) is widely used to reduce energy consumption. Recovery of heat from drying air is more difficult than from steam or vapours because air is not recompressible. However, a number of heat exchanger designs are used to recover waste heat from air or gases (for example the 'thermal wheel' in Fig. 1.11).

Commercial applications of this equipment are described in Chapter 15 (baking ovens) and Chapter 16 (deep-fat fryers). If the humidity of air is reduced, it is reheated more economically. The 'rotaire wheel' (Fig. 1.12) and heat pump are examples of dehumidifying equipment. Heat pumps are similar to refrigeration plant (see Chapter 19, Fig. 19.1) but operate by removing heat from a low-temperature source and using it to heat air or water. The application of heat pumps to low-temperature drying of

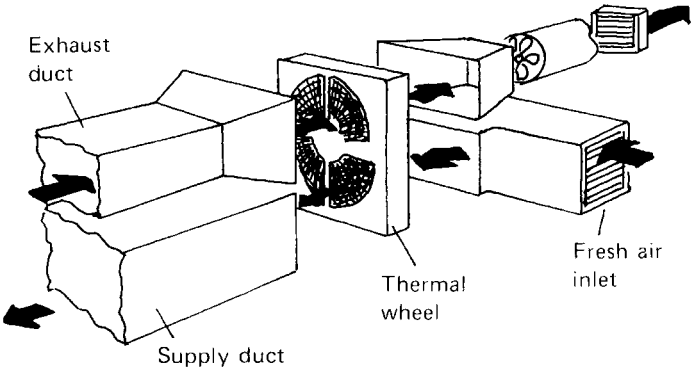


Fig. 1.11 Thermal wheel (Courtesy of the Electricity Council.)

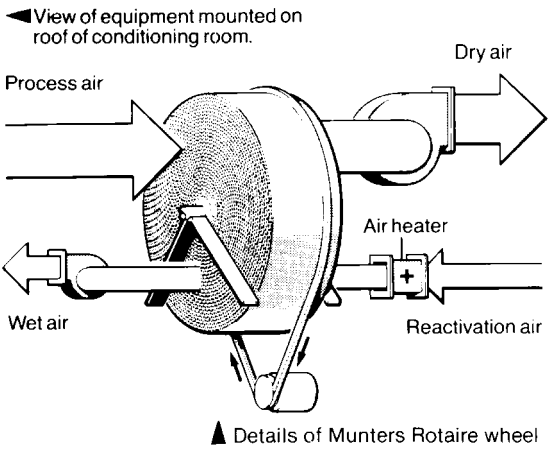


Fig 1.12 Rotaire wheel (Courtesy of the Electricity Council.)

confectionery is described by Anon. (1983a, b, c). The use of air knives for package drying is described by Anon. (1983d) and Beevers (1985). Other energy-saving techniques during dehydration are described by Senhaji and Bimbenet (1984), Flink (1977) and Green (1982).

1.4.5 Effect of heat on micro-organisms

The preservative effect of heat processing is due to the denaturation of proteins, which destroys enzyme activity and enzyme-controlled metabolism in micro-organisms. The rate of destruction is a first-order reaction; that is when food is heated to a temperature that is high enough to destroy contaminating micro-organisms, the same percentage die in a given time interval regardless of the numbers present initially. This is known as the *logarithmic order of death* and is described by a *death rate curve* (Fig. 1.13).

The time needed to destroy 90% of the micro-organisms (to reduce their numbers by a factor of 10) is referred to as the *decimal reduction time* or *D* value (5 min in Fig. 1.13). *D* values differ for different microbial species (Table 1.11), and a higher *D* value indicates greater heat resistance.

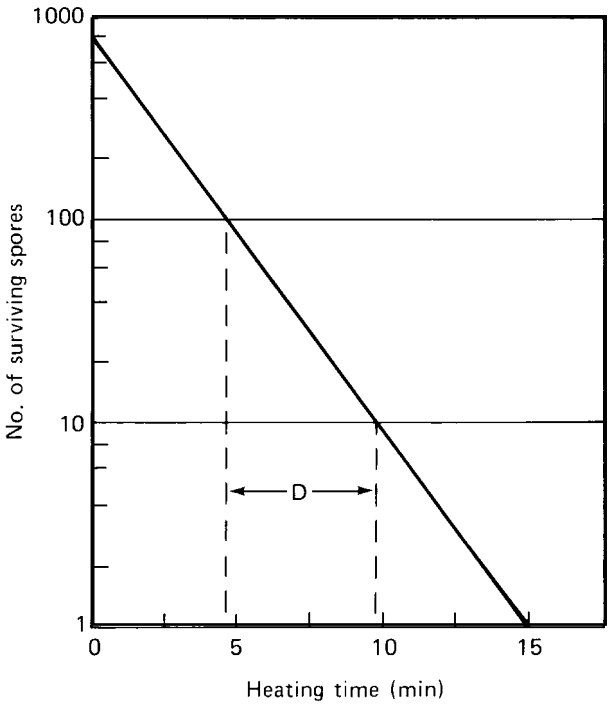


Fig. 1.13 Death rate curve.

Table 1.11 Thermal properties of selected nutritional and sensory components of foods in relation to heat-resistant enzymes and bacteria

Component	Source	pH	z (°C)	D_{121} (min)	Temperature range (°C)
Thiamin	Carrot purée	5.9	25	158	109–149
Thiamin	Pea purée	Natural	27	247	121–138
Thiamin	Lamb purée	6.2	25	120	109–149
Lysine	Soya bean meal	–	21	786	100–127
Chlorophyll a	Spinach	6.5	51	13.0	127–149
Chlorophyll a	Spinach	Natural	45	34.1	100–130
Chlorophyll b	Spinach	5.5	79	14.7	127–149
Chlorophyll b	Spinach	Natural	59	48	100–130
Anthocyanin	Grape juice	Natural	23.2	17.8*	20–121
Betanin	Beetroot juice	5.0	58.9	46.6*	50–100
Carotenoids	Paprika	Natural	18.9	0.038*	52–65
Peroxidase	Peas	Natural	37.2	3.0	110–138
Peroxidase	Various	–	28–44	–	–
Clostridium botulinum spores type A + B	Various	>4.5	5.5–10	0.1–0.3*	104
Bacillus stereotherm-ophilus	Various	>4.5	7–12	4.0–5.0	110+

* *D* values at temperatures other than 121°C.
Adapted from Felliciotti and Esselen (1957), Taira *et al.* (1966), Gupta *et al.* (1964), Ponting *et al.* (1960), von Elbe *et al.* (1974), Adams and Yawger (1961), Esselen and Anderson (1956) and Stumbo (1973).

There are two important implications arising from the decimal reduction time: first, the higher the number of micro-organisms present in a raw material, the longer it takes to reduce the numbers to a specified level. In commercial operation the number of micro-organisms varies in each batch of raw material, but it is difficult to recalculate process times for each batch of food. A specific temperature–time combination is therefore used to process every batch of a particular product, and adequate preparation procedures (Chapter 3) are used to ensure that the raw material has a satisfactory and uniform microbiological quality. Second, because microbial destruction takes place logarithmically, it is theoretically possible to destroy all cells only after heating for an infinite time. Processing therefore aims to reduce the number of surviving micro-organisms by a pre-determined amount. This gives rise to the concept of *commercial sterility*, which is discussed further in Chapters 10–12.

The destruction of micro-organisms is temperature dependent; cells die more rapidly at higher temperatures. By collating *D* values at different temperatures, a *thermal death time* (TDT) curve is constructed (Fig. 1.14). The slope of the TDT curve is termed the *z* value and is defined as the number of degrees Celsius required to bring about a ten-fold change in decimal reduction time (10.5ºC in Fig. 1.14). The *D* value and *z* value are used to characterise the heat resistance of a micro-organism and its temperature dependence respectively.

There are a large number of factors which determine the heat resistance of micro-organisms, but general statements of the effect of a given variable on heat resistance are not always possible. The following factors are known to be important.

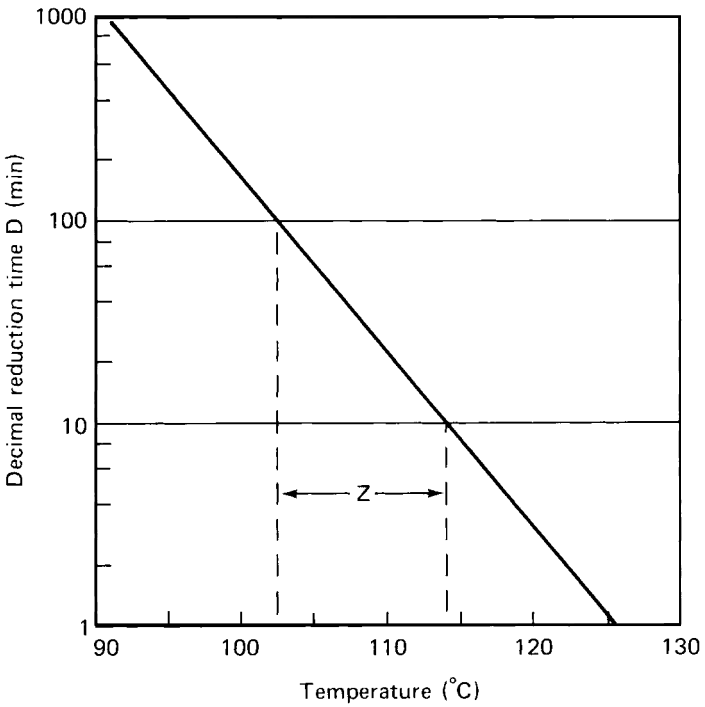


Fig. 1.14 TDT curve. Microbial destruction is faster at higher temperatures (for example 100 min at 102.5ºC has the same lethal effect as 10 min at 113ºC).

1. *Type of micro-organism*. Different species and strains show wide variation in their heat resistance (Table 1.11). Spores are much more heat resistant than vegetative cells.
2. *Incubation conditions* during cell growth or spore formation. These include:
   (a) temperature (spores produced at higher temperatures are more resistant than those produced at lower temperatures)
   (b) age of the culture (the stage of growth of vegetative cells affects their heat resistance)
   (c) culture medium used (for example mineral salts and fatty acids influence the heat resistance of spores).
3. *Conditions during heat treatment*. The important conditions are:
   (a) pH of the food (pathogenic and spoilage bacteria are more heat resistant near to neutrality; yeasts and fungi are able to tolerate more acidic conditions but are less heat resistant than bacterial spores)
   (b) water activity of the food (section 1.5) influences the heat resistance of vegetative cells; in addition moist heat is more effective than dry heat for spore destruction
   (c) composition of the food (proteins, fats and high concentration of sucrose increase the heat resistance of micro-organisms; the low concentration of sodium chloride used in most foods does not have a significant effect; the physical state of the food, particularly the presence of colloids, affects the heat resistance of vegetative cells)
   (d) the growth media and incubation conditions, used to assess recovery of micro-organisms in heat resistance studies, affect the number of survivors observed.

Most enzymes have *D* and *z* values within a similar range to micro-organisms, and are therefore inactivated during normal heat processing. However, some enzymes are very heat resistant. These are particularly important in acidic foods, where they may not be completely denaturated by the relatively short heat treatments and lower temperatures required for microbial destruction. The factors which influence heat resistance of enzymes are similar to those described for micro-organisms and are discussed in detail by Whitaker (1972).

A knowledge of the heat resistance of the enzymes and/or micro-organisms found in a specific food is used to calculate the heating conditions needed for their destruction. In practice the most heat resistant enzyme or micro-organism likely to be present in a given food is used as a basis for calculating process conditions. It is assumed that other less heat-resistant species are also destroyed. Methods for the calculation of processing time are described in Chapter 12.

1.4.6 Effect of heat on nutritional and sensory characteristics

The destruction of many vitamins, aroma compounds and pigments by heat follows a similar first-order reaction to microbial destruction. Examples of *D* and *z* values of selected vitamins and pigments are shown in Table 1.11. In general both values are higher than those of micro-organisms and enzymes. As a result, nutritional and sensory properties are better retained by the use of higher temperatures and shorter times during heat processing. It is therefore possible to select particular time-temperature combinations from a TDT curve (all of which achieve the same degree of enzyme or microbial destruction), to optimise a process for nutrient retention or preservation of desirable sensory qualities. This

concept forms the basis of individual quick blanching (Chapter 10), high-temperature short-time (HTST) pasteurisation (Chapter 11), ultrahigh-temperature sterilisation (Chapter 12) and HTST extrusion (Chapter 14). The loss of nutrients and changes to sensory quality during heat processing are reported in Chapters 10–18.

1.5 Water activity

Deterioration of foods by micro-organisms can take place rapidly, whereas enzymic and chemical reactions take place more slowly during storage. In either case the water content is a very important factor controlling the rate of deterioration. The moisture content of foods can be expressed either on a wet-weight basis:

$$m = \frac{\text{mass of water}}{\text{mass of sample}} \times 100 \qquad 1.28$$

$$m = \frac{\text{mass of water}}{\text{mass of water} + \text{solids}} \times 100 \qquad 1.29$$

or on a dry-weight basis

$$m = \frac{\text{mass of water}}{\text{mass of solids}} \qquad 1.30$$

(Lewis, 1990).

The dry-weight basis is more commonly used for processing calculations, whereas the wet-weight basis is frequently quoted in food composition tables. It is important, however, to note which system is used when expressing a result. Wet-weight basis is used throughout this text unless otherwise stated.

A knowledge of the moisture content alone is not sufficient to predict the stability of foods. Some foods are unstable at a low moisture content (for example peanut oil deteriorates if the moisture content exceeds 0.6%), whereas other foods are stable at relatively high moisture contents (for example potato starch is stable at 20% moisture) (van den Berg, 1986). It is the *availability* of water for microbial, enzymic or chemical activity that determines the shelf life of a food, and this is measured by the water activity (a_w) of a food, also known as the Relative Vapour Pressure (RVP).

Examples of unit operations that reduce the availability of water in foods include those that physically remove water (dehydration (Chapter 15), evaporation (Chapter 13) and freeze drying or freeze concentration (Chapter 22)) and those that immobilise water in the food (for example by the use of humectants in 'intermediate-moisture' foods and by formation of ice crystals in freezing (Chapter 21)). Examples of the moisture content and a_w of foods are shown in Table 1.12 and the effect of reduced a_w on food stability is shown in Table 1.13.

Water in food exerts a vapour pressure. The size of the vapour pressure depends on:

- the amount of water present
- the temperature
- the concentration of dissolved solutes (particularly salts and sugars) in the water.

Water activity is defined as 'the ratio of the vapour pressure of water in a food to the saturated vapour pressure of water at the same temperature' (equation 1.31):

Table 1.12 Moisture content and water activity of foods

Food	Moisture content (%)	Water activity	Degree of protection required
Ice (0°C)	100	1.00[a]	
Fresh meat	70	0.985	
Bread	40	0.96	Package to prevent moisture loss
Ice (−10°C)	100	0.91[a]	
Marmalade	35	0.86	
Ice (−20°C)	100	0.82[a]	
Wheat flour	14.5	0.72	
Ice (−50°C)	100	0.62[a]	Minimum protection or no packaging required
Raisins	27	0.60	
Macaroni	10	0.45	
Cocoa powder		0.40	
Boiled sweets	3.0	0.30	
Biscuits	5.0	0.20	Package to prevent moisture uptake
Dried milk	3.5	0.11	
Potato crisps	1.5	0.08	

[a] Vapour pressure of ice divided by vapour pressure of water.
Adapted from Troller and Christian (1978), van den Berg (1986) and Brenndorfer *et al*. (1985).

Table 1.13 The importance of water activity in foods

a_w	Phenomenon	Examples
1.00		Highly perishable fresh foods
0.95	*Pseudomonas*, *Bacillus*, *Clostridium perfringens* and some yeasts inhibited	Foods with 40% sucrose or 7% salt, cooked sausages, bread
0.90	Lower limit for bacterial growth (general), *Salmonella*, *Vibrio parahaemolyticus*, *Clostridium botulinum*, *Lactobacillus*, and some yeasts and fungi inhibited	Foods with 55% sucrose, 12% salt, cured ham, medium-age chesse. Intermediate-moisture foods ($a_w = 0.90 - 0.55$)
0.85	Many yeasts inhibited	Foods with 65% sucrose, 15% salt, salami, mature cheese, margarine
0.80	Lower limit for most enzyme activity and growth of most fungi; *Straphlococcus aureus* inhibited	Flour, rice (15–17% water) fruit cake, sweetened condensed milk, fruit syrups, fondant
0.75	Lower limit for halophilic bacteria	Marzipan (15–17% water), jams
0.70	Lower limit for growth of most xerophilic fungi	
0.65	Maximum velocity of Maillard reactions	Rolled oats (10% water), fudge, molasses, nuts
0.60	Lower limit for growth of osmophilic or xerophilic yeasts and fungi	Dried fruits (15–20% water), toffees, caramels (8% water), honey
0.55	Deoxyribonucleic acid becomes disordered (lower limit for life to continue)	
0.50		Dried foods ($a_w = 0$–0.55), spices, noodles
0.40	Minimum oxidation velocity	Whole egg powder (5% water)
0.30		Crackers, bread crusts (3–5% water)
0.25	Maximum heat resistance of bacterial spores	
0.20		Whole milk powder (2–3% water), dried vegetables (5% water), cornflakes (5% water)

$$a_w = \frac{P}{P_0} \quad [1.31]$$

where P (Pa) = vapour pressure of the food, P_0 (Pa) = vapour pressure of pure water at the same temperature. a_w is related to the moisture content by a number of equations, including the following:

$$\frac{a_w}{M(1-a_w)} = \frac{1}{M_1 C} + \frac{C-1}{M_1 C} a_w \quad [1.32]$$

where a_w = water activity, M = moisture as percentage dry weight, M_1 = moisture (dry-weight basis) of a monomolecular layer and C = a constant. A sample calculation is described by Karel (1975).

A proportion of the total water in a food is strongly bound to specific sites (for example hydroxyl groups of polysaccharides, carbonyl and amino groups of proteins, and hydrogen bonding). When all sites are (statistically) occupied by adsorbed water the moisture content is termed the *BET* (Brunauer–Emmett–Teller) *monolayer* value (Fennema, 1996). Typical examples include gelatin (11%), starch (11%), amorphous lactose (6%) and whole spray-dried milk (3%). The BET monolayer value therefore represents the moisture content at which the food is most stable. At moisture contents below this level, there is a higher rate of lipid oxidation and, at higher moisture contents, Maillard browning and then enzymic and microbiological activities are promoted (Fig. 1.15).

The movement of water vapour from a food to the surrounding air depends upon both the food (moisture content and composition) and the condition of the air (temperature and humidity). At a constant temperature, the moisture content of food changes until it comes into equilibrium with water vapour in the surrounding air. The food then neither gains nor loses weight on storage under those conditions. This is called the *equilibrium moisture content* of the food and the relative humidity of the storage atmosphere is known as the *equilibrium relative humidity*. When different values of relative humidity versus equilibrium moisture content are plotted, a curve known as a *water sorption isotherm* is obtained (Fig. 1.16).

Each food has a unique set of sorption isotherms at different temperatures. The precise shape of the sorption isotherm is caused by differences in the physical structure, chemical composition and extent of water binding within the food, but all sorption isotherms have a characteristic shape, similar to that shown in Fig. 1.16. The first part of the curve, to point A, represents monolayer water, which is very stable, unfreezable and not removed by drying. The second, relatively straight part of the curve (AB) represents water adsorbed in multilayers within the food and solutions of soluble components. The third portion, (above point B) is 'free' water condensed within the capillary structure or in the cells of a food. It is mechanically trapped within the food and held by only weak forces. It is easily removed by drying and easily frozen, as indicated by the steepness of the curve. Free water is available for microbial growth and enzyme activity, and a food which has a moisture content above point B on the curve is likely to be susceptible to spoilage.

The sorption isotherm indicates the a_w at which a food is stable and allows predictions of the effect of changes in moisture content on a_w and hence on storage stability. It is used to determine the rate and extent of drying (Chapter 15), the optimum frozen storage temperatures (Chapter 21) and the moisture barrier properties required in packaging materials (Chapter 24).

The rate of change in a_w on a sorption isotherm differs according to whether moisture is removed from a food (*desorption*) or whether it is added to dry food (*absorption*) (Fig. 1.16). This is termed a *hysteresis loop*. The difference is large in some foods (for example

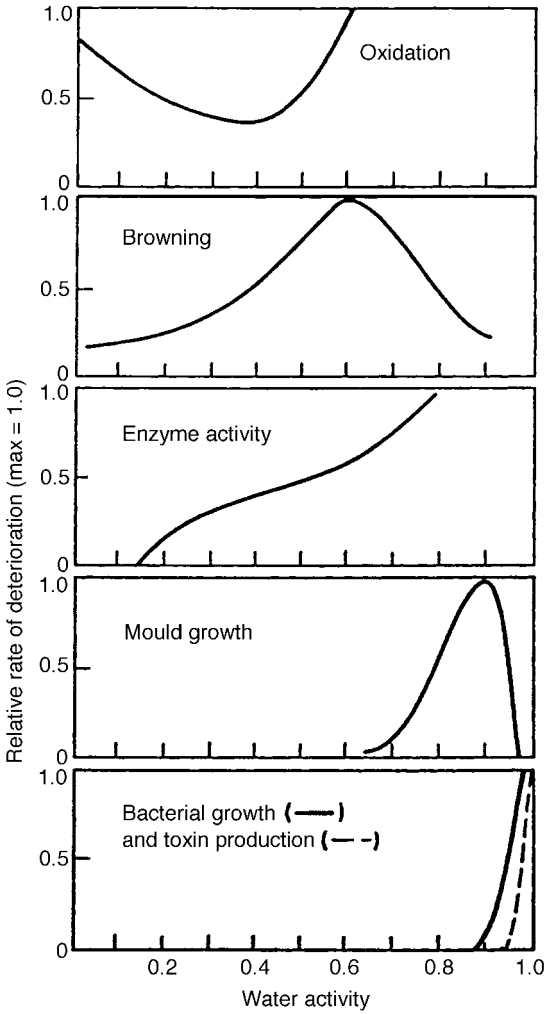


Fig. 1.15 Effect of water activity on microbial, enzymic and chemical changes to foods. (After Karel (1975).)

rice) and is important for example in determining the protection required against moisture uptake.

1.5.1 Effect of a_w on foods

The effect of a_w on microbiological and selected biochemical reactions is shown in Fig. 1.15 and Table 1.13. Almost all microbial activity is inhibited below $a_w = 0.6$, most fungi are inhibited below $a_w = 0.7$, most yeasts are inhibited below $a_w = 0.8$ and most bacteria below $a_w = 0.9$. The interaction of a_w with temperature, pH, oxygen and carbon dioxide, or chemical preservatives has an important effect on the inhibition of microbial growth. When any one of the other environmental conditions is sub-optimal for a given micro-organism, the effect of reduced a_w is enhanced. This permits the combination of several mild control mechanisms which result in the preservation of food without substantial loss of nutritional value or sensory characteristics (Table 1.14) (see also the 'Hurdle effect' in Section 1.8.2).

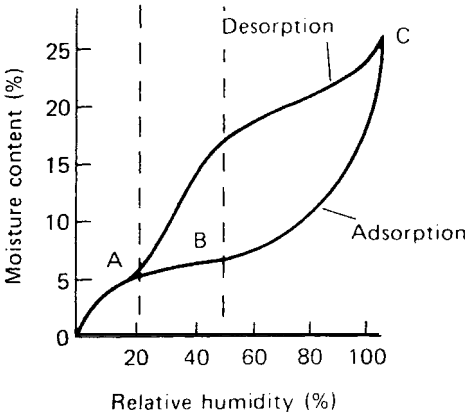


Fig. 1.16 Water sorption isotherm.

Table 1.14 Interaction of a_w, pH and temperature in selected foods

Food	pH	a_w	Shelf life	Notes
Fresh meat	>4.5	>0.95	days	Preserve by chilling
Cooked meat	>4.5	0.95	weeks	Ambient storage when packaged
Dry sausage	>4.5	<0.90	months	Preserved by salt and low a_w
Fresh vegetables	>4.5	>0.95	weeks	'Stable' while respiring
Pickles	<4.5	0.90	months	Low pH maintained by packaging
Bread	>4.5	>0.95	days	
Fruit cake	>4.5	<0.90	weeks	Preserved by heat and low a_w
Milk	>4.5	>0.95	days	Preserved by chilling
Yoghurt	<4.5	<0.95	weeks	Preserved by low pH and chilling
Dried milk	>4.5	<0.90	months	Preserved by low a_w

Enzymic activity virtually ceases at a_w values below the BET monolayer value. This is due to the low substrate mobility and its inability to diffuse to the reactive site on the enzyme. Chemical changes are more complex. The two most important that occur in foods which have a low a_w are Maillard browning and oxidation of lipids. The a_w that causes the maximum rate of browning varies with different foods. However, in general, a low a_w restricts the mobility of the reactants and browning is reduced. At a higher a_w, browning reaches a maximum. Water is a product of the condensation reaction in browning and, at higher moisture levels, browning is slowed by 'end product inhibition'. At high moisture contents, water dilutes the reactants and the rate of browning falls.

Oxidation of lipids occurs at low a_w values owing to the action of free radicals. Above the BET monolayer value, anti-oxidants and chelating agents (which sequester trace metal catalysts) become soluble and reduce the rate of oxidation. At higher a_w values the catalytic activity of metals is reduced by hydration and the formation of insoluble hydroxides but, at high a_w values, metal catalysts become soluble, and the structure of the food swells to expose more reactive sites (Fig. 1.15).

1.6 Effects of processing on the sensory characteristics of foods

There are a number of definitions of 'quality' of foods, which are discussed by Cardello (1998). To the consumer, the most important quality attributes of a food are its sensory

characteristics (texture, flavour, aroma, shape and colour). These determine an individual's preference for specific products, and small differences between brands of similar products can have a substantial influence on acceptability. A continuing aim of food manufacturers is to find improvements in processing technology which retain or create desirable sensory qualities or reduce the damage to food caused by processing. Examples of such improvements are described in subsequent chapters.

1.6.1 Texture

The texture of foods is mostly determined by the moisture and fat contents, and the types and amounts of structural carbohydrates (cellulose, starches and pectic materials) and proteins that are present. Changes in texture are caused by loss of moisture or fat, formation or breakdown of emulsions and gels, hydrolysis of polymeric carbohydrates, and coagulation or hydrolysis of proteins. Detailed information on the textural characteristics of food is given by Mohsenin (1980), Lewis (1990) and Szczesniak (1983). The effect of food composition and structure on texture is described by Stanley and Tung (1976) and Sherman (1976).

1.6.2 Taste, flavour and aroma

Taste attributes consist of saltiness, sweetness, bitterness and acidity and some of these attributes can be detected in very low thresholds in foods (Table 1.15). The taste of foods is largely determined by the formulation used for a particular food and is mostly unaffected by processing. Exceptions to this include increased sweetness due to

Table 1.15 Detection thresholds for common food components

Compound	Taste or odour	Threshold
Taste compounds		
Hydrochloric acid	Sour	0.0009 N
Citric acid	Sour	0.0016 N
Lactic acid	Sour	0.0023 N
Sodium chloride	Salty	0.01 M
Potassium chloride	Bitter/salty	0.017 M
Sucrose	Sweet	0.01 M
Glucose	Sweet	0.08 M
Sodium saccharine	Sweet	0.000023 M
Quinine sulphate	Bitter	0.000008 M
Caffeine	Bitter	0.0007 M
Odour compounds		
Citral	Lemon	0.000003 mg/l
Limonene	Lemon	0.1 mg/l
Butyric acid	Rancid butter	0.009 mg/l
Benzaldehyde	Bitter almond	0.003 mg/l
Ethyl acetate	Fruity	0.0036 mg/l
Methyl salicylate	Wintergreen	0.1 mg/l
Hydrogen sulphide	Rotten eggs	0.00018 mg/l
Amyl acetate	Banana oil	0.039 mg/l
Safrol	Sassafras	0.005 mg/l
Ethyl mercaptan	Rotten cabbage	0.00000066 mg/l

From Cardello (1998).

respiratory changes in fresh foods (Chapter 19) and changes in acidity or sweetness during food fermentations (Chapter 7). Fresh foods contain complex mixtures of volatile compounds, which give characteristic flavours and aromas, some of which are detectable at extremely low concentrations (Table 1.15). These compounds may be lost during processing, which reduces the intensity of flavour or reveals other flavour/aroma compounds. Volatile aroma compounds are also produced by the action of heat, ionising radiation, oxidation or enzyme activity on proteins, fats and carbohydrates. Examples include the Maillard reaction between amino acids and reducing sugars or carbonyl groups and the products of lipid degradation (Chapter 16), or hydrolysis of lipids to fatty acids and subsequent conversion to aldehydes, esters and alcohols. The perceived aroma of foods arises from complex combinations of many hundreds of compounds, some of which act synergistically (Maruniak and MacKay-Sim, 1984). In addition, the perceived flavour of foods is influenced by the rate at which flavour compounds are released during chewing, and hence is closely associated with the texture of foods and the rate of breakdown of food structure during mastication (Clark, 1990). Research into flavour release is described by Wyeth and Kilcast (1991) and Plattig (1984) and more detailed descriptions of the production of aroma compounds or loss of naturally occurring aroma compounds are given by Lindsay (1996) and in subsequent chapters.

1.6.3 Colour

Methods to assess the colour and appearance of foods are described by MacDougall (1984). Many naturally occurring pigments are destroyed by heat processing, chemically altered by changes in pH or oxidised during storage (Table 1.16). As a result the processed food may lose its characteristic colour and hence its value. Synthetic pigments (Appendix C) are more stable to heat, light and changes in pH, and they are therefore added to retain the colour of some processed foods. Details of changes to naturally occurring pigments are described in Chapters 10–17. Maillard browning is an important cause of both desirable changes in food colour (for example in baking or frying (Chapters 16 and 17), and in the development of off-colours (for example during canning and drying (Chapters 12 and 15)).

1.7 Effects of processing on nutritional properties

Many unit operations, especially those that do not involve heat, have little or no effect on the nutritional quality of foods. Examples include mixing, cleaning, sorting, freeze drying and pasteurisation. Unit operations that intentionally separate the components of foods (Chapters 3, 4 and 6) alter the nutritional quality of each fraction compared with the raw material. Unintentional separation of water-soluble nutrients (minerals, water-soluble vitamins and sugars) also occurs in some unit operations (for example blanching (Chapter 10), and in drip losses from roast or frozen foods (Chapters 16 and 21)).

Heat processing is a major cause of changes to nutritional properties of foods. For example gelatinisation of starches and coagulation of proteins improve their digestibility, and anti-nutritional compounds (for example a trypsin inhibitor in legumes) are destroyed. However, heat also destroys some types of heat-labile vitamin (Fig. 1.17), reduces the biological value of proteins (owing to destruction of amino acids or Maillard browning reactions) and promotes lipid oxidation. The effects of heat on proteins and lipids are described in detail by Mauron (1982) and Witting and Dimick (1982) respectively.

Table 1.16 Naturally occurring pigments in foods

Pigment	Typical source	Oil or water soluble	Stability to the following			
			Heat	Light	Oxygen	pH change
Anthocyanins	Fruits	Water soluble	High	High	High	Low
Betalaines	Beetroot	Water soluble	Moderate	High	High	High
Bixin	Seed coat of *Bixa orellana*	Oil soluble	Moderate to low	Low	High	–
Canxanthin		Oil soluble	Moderate	Moderate	Moderate	Moderate
Caramel	Heated sugar	Water soluble	High	High	High	High
Carotenes	Leaves	Oil soluble	Moderate to low	Low	Low	High
Chlorophylls	Leaves	Water soluble	High	High	High	Low
Cochineal[a]	Insect (*Dactylopius coccus*)	Water soluble	High	High	–	Moderate to high
Curcumin	Turmeric	Water soluble	Low	Low	Low	–
Norbixin	See Bixin	Water soluble	Moderate to low	Low	High	–
Oxymyoglobin	Animals	Water soluble	Low	–	High	Low
Polyphenols	Tea leaf	Water soluble	High	High	High	High
Quinones	Roots, bark	Water soluble	High	Moderate	–	Moderate
Xanthophylls	Fruits	Water soluble	Moderate	High	High	Low

[a] As aluminium lake.
From the data of Zapsalis and Beck (1985) and Coultate (1984).

Oxidation is a second important cause of nutritional changes to foods. This occurs when food is exposed to air (for example in size reduction or hot-air drying (Chapters 4 and 15)) or as a result of the action of heat or oxidative enzymes (for example peroxidase or lipoxygenase). The main nutritional effects of oxidation are:

- the degeneration of lipids to hydroperoxides and subsequent reactions to form a wide variety of carbonyl compounds, hydroxy compounds and short chain fatty acids, and in frying oils to toxic compounds (Chapter 17)
- the destruction of oxygen-sensitive vitamins (Fig. 1.17 and Appendix A).

Changes to lipids are discussed by Zapsalis and Beck (1985) and a discussion of the properties of vitamins and losses during processing is given by Bender (1978), Harris and Karmas (1975) and De Ritter (1982). Examples of vitamin losses caused by individual unit operations are described in subsequent chapters.

The importance of nutrient losses during processing depends on the nutritional value of a particular food in the diet. Some foods (for example bread, potatoes and milk in Western countries and rice or maize in Eastern and developing countries) are an

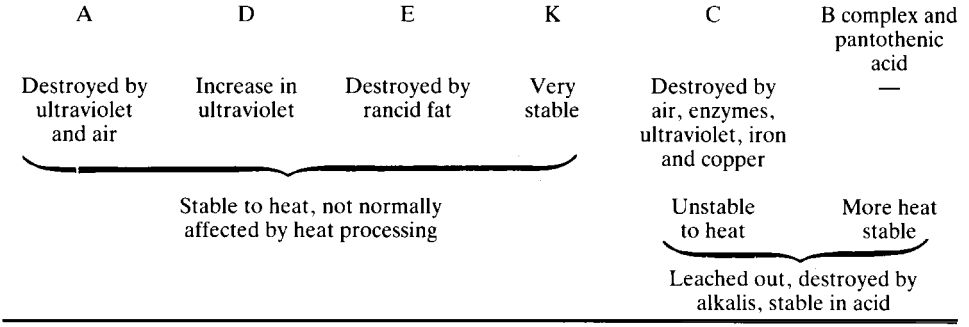


Fig. 1.17 Stability of vitamins in foods.

important source of nutrients for large numbers of people. Vitamin losses are therefore more significant in these foods than in those which either are eaten in small quantities or have low concentrations of nutrients.

In industrialised countries, the majority of the population achieve an adequate supply of nutrients from the mixture of foods that is eaten. Losses due to processing of one component of the diet are therefore insignificant to the long-term health of an individual. In an example described by Bender (1987), complete meals which initially contained 16.5 μg of vitamin A lost 50% on canning and 100% after storage for 18 months. Although the losses appear to be significant, the original meal contained only 2% of the recommended daily intake (RDA), and the extent of loss is therefore of minor importance. The same meal contained 9 mg of thiamin and lost 75% after 18 months' storage. The thiamin content is ten times the RDA, and adequate quantities therefore remained. Possible exceptions are the special dietary needs of pre-term infants, pregnant women and the elderly. In these groups there may be either a special need for certain nutrients or a more restricted diet than normal. These special cases are discussed in detail by Watson (1986) and Francis (1986).

Reported vitamin losses during processing are included in subsequent chapters to give an indication of the severity of each unit operation. However, such data should be treated with caution. Variation in nutrient losses between cultivars or varieties can exceed differences caused by alternative methods of processing. Growth conditions, or handling and preparation procedures prior to processing, also cause substantial variation in nutrient loss. Data on nutritional changes cannot be directly applied to individual commercial operations, because of differences in ingredients, processing conditions and equipment used by different manufacturers.

1.8 Food safety, good manufacturing practice and quality assurance

Traditionally, quality control systems were based on the inspection of a product at various points within a process, and rejection of any product that did not meet agreed standards. This reactive approach to food quality focused mainly on end-product testing, which is now recognised to be a waste of resources (resources have already been spent on producing the food by the time it is tested and rejection means a financial loss). A more proactive preventative approach to food safety and quality management, termed 'Quality Assurance' was developed during the 1980s, based on the principles of Good Manufacturing Practice (GMP) (Anon., 1998). It aimed to ensure that quality and safety

are maintained throughout a process and thus prevent product rejection and financial losses.

The pressures for this development came from two sources: first, commercial pressures including:

- increasing competition between companies
- the need to access expanding national and international food markets
- product quality management systems required by major retailers.

These all required an effective food monitoring and control system to control resources and ensure that safe, high quality products were consistently manufactured, and led to the concept of Total Quality Management (TQM). Second, new legislation demanded systems that would both maintain quality and safety, and prove that a business is in control of these. The initial quality assurance legislation (BS 5750/BS EN ISO 9000 standards) has now been expanded to take all business operations into account. The aim of TQM is therefore for companies to define and understand *all* of their processes, to implement controls, monitor performance and measure improvements. The standard BS 7850 defines TQM as 'management philosophy and company practices that aim to harness the human and material resources of an organisation in the most effective way to achieve the objectives of the organisation'. This concept takes into account separate standards for production operations and new product development (BS EN ISO 9001), production facilities (BS EN ISO 9002), environmental and waste management (ISO 14001, also Chapter 26), employment rights, health and safety (BS 8800), laboratory management (EN 45000) as well as food quality and safety.

As part of a TQM programme, food manufacturers must also take into account the requirements of a food safety management system termed 'HACCP' (Hazard Analysis Critical Control Point), which was developed by Codex Alimentarius. This provides the key control measures needed to understand the mechanisms for producing safe food and the basis to create production control systems to ensure product quality. A major shift in emphasis from national legislation to international legislation occurred in 1994, when a GATT[2] agreement recommended acceptance of HACCP as the required standard for free international movement of food. The WTO now evaluates both technical barriers to trade and sanitary controls with reference to Codex HACCP standards (Mitchell, 1996).

More recently in the UK, the British Retail Consortium has issued a 'Technical Standard for Companies Supplying Retailer Branded Food Products', which is being used by retailers as the definitive standard for suppliers, and forms the basis of their terms of business (Rose, 2000). It covers six key areas: HACCP systems, quality management systems, factory environmental standards, product control, process control and personnel, and is implemented through third-party inspection bodies (e.g. the European Food Safety Inspection Service (EFSIS)).

TQM is therefore broader than quality control, and is a management philosophy that seeks to continually improve the effectiveness and competitiveness of the business as a whole. It is an integrated system which ensures that all areas of the business are controlled, to enable customers to consistently receive quality products that meet their needs and expectations. The approach requires collective responsibility and commitment at all levels in a business, and can only be achieved through trusting working relationships and good communications. Quality management systems are usually developed with reference to an international standard or specific legislative standards

2. General Agreement on Tariffs and Trade, later replaced by the World Trade Organisation (WTO).

(e.g. 91/493 European standard for fisheries products). However, any reference standard is only a framework on which a company builds its own performance standards to meet the requirements of the law, the customers and the company's own requirements.

In outline, a TQM system covers the following areas:

- raw materials, purchasing and control (including agreed specifications, supplier auditing, raw material storage, stock control, traceability, inspection, investigation of non-conformity to specification)
- process control (including identification, verification and monitoring of critical control points in a HACCP scheme, hygienic design of plant and layouts to minimise cross-contamination, cleaning schedules, recording of critical production data, sampling procedures and contingency plans to cover safety issues)
- premises (including methods of construction to minimise contamination, maintenance, waste disposal)
- quality control (including product specifications and quality standards for non-safety quality issues, monitoring and verification of quality before distribution)
- personnel (including training, personal hygiene, clothing and medical screening)
- final product (including types and levels of inspection to determine conformity with quality specifications, isolating non-conforming products, packaging checks, inspection records, complaints monitoring systems)
- distribution (to maintain the product integrity throughout the chain, batch traceability and product recall systems).

Mann and Adebanjo (1997) and Rose (2000) have produced overviews of TQM, and Dillon (1999) describes a practical approach to beginning TQM projects in smaller businesses.

A TQM system requires a company to have a level of organisation and management that enables it to meet the TQM objectives and this requires the company to:

- have a defined organisational structure
- clearly allocate authority and division of responsibilities
- define procedures
- link the components of the system
- prepare key documentation
- allocate sufficient resources for implementation of the system (Dillon and Griffith, 1996).

Rose (2000) has emphasised the need for commitment by senior management to implementation of a TQM system and the need for senior managers to review and update the effectiveness of systems that are in place. The component procedures for introducing a quality management system include:

- development of audit methods
- procedures for corrective action
- procedures for management review and
- documentation of the system.

Effective implementation requires the combined use of all resources, including people, and Dillon (1996) has emphasised that many systems fail because inadequate attention is paid to the staff who will implement the system. *Audits* are the regular systematic collection and feedback of objective information by competent independent personnel. They are used for example to monitor suppliers' ability to meet agreed

requirements or to monitor production routines. They are an effective tool for monitoring the success of a quality system and are described in detail by Sayle (1998), Mortimer and Wallace (1998) and Dillon and Griffith (1997). The types of *non-conformance* with specifications, performance standards, customer requirements or legislation for a particular operation in a factory department must be clearly defined and appropriate action agreed if a failure occurs. For example, if deviation is a legal issue, the appropriate action is to quarantine the product, or if it is deviation from a customer requirement, the action may involve informing customer and sale at reduced price.

Management review is an essential component of any system and managers must consider the feedback from the system and decide on appropriate action, using formal meetings with standard agendas, actions and responsibilities. The need to manage information by *document control* ensures that documents are approved, issued and stored in agreed places in filing cabinets, computer diskettes or hard drives or on CD. Agreed reference codes, directory structures, storage procedures, archiving instructions and backup mechanisms are essential. There is a hierarchy of documents, from manuals outlining policy, procedures that link key activities, detailed work instructions and output records which requires an organisation to prove that a system gives the necessary control. Document control rules therefore need to be agreed and put in place early in the development of a control system. Procedures for *corrective action* are needed to prevent a recurrence of any deficiencies within the system.

The benefits of a properly implemented TQM system are stated by Rose (2000) as:

- economic (more cost-effective production by 'getting it right first time', reduction in wasted materials, fewer customer complaints, improved machine efficiency and increased manufacturing capacity)
- marketing (consistently meeting customer needs, increased customer confidence and sales)
- internal (improved staff morale, increased levels of communication, better trained staff and awareness/commitment to quality, improved management control and confidence in the operation)
- legislative (demonstrating due diligence, providing evidence of commitment to quality and ability to improve).

1.8.1 HACCP

Hazard analysis is 'the identification of potentially hazardous ingredients, storage conditions, packaging, critical process points and relevant human factors which may affect product safety or quality'. HACCP enables potential hazards in a process to be identified, assessed, and controlled or eliminated. It sets tolerances for hazards and defines appropriate control measures, the frequency of their application, sampling procedures, specific tests to be used and the criteria for product acceptance. The system is based on monitoring of *critical control points*[3] and the action to be taken when results of monitoring are outside the pre-set limits (Fig 1.18). HACCP is used throughout each stage of a process, and includes raw materials, processing, storage and distribution. It can be used for all potential hazards, including inadequate quality as well as safety, and can

3. These are processing factors of which a loss of control would result in an unacceptable food safety or quality risk.

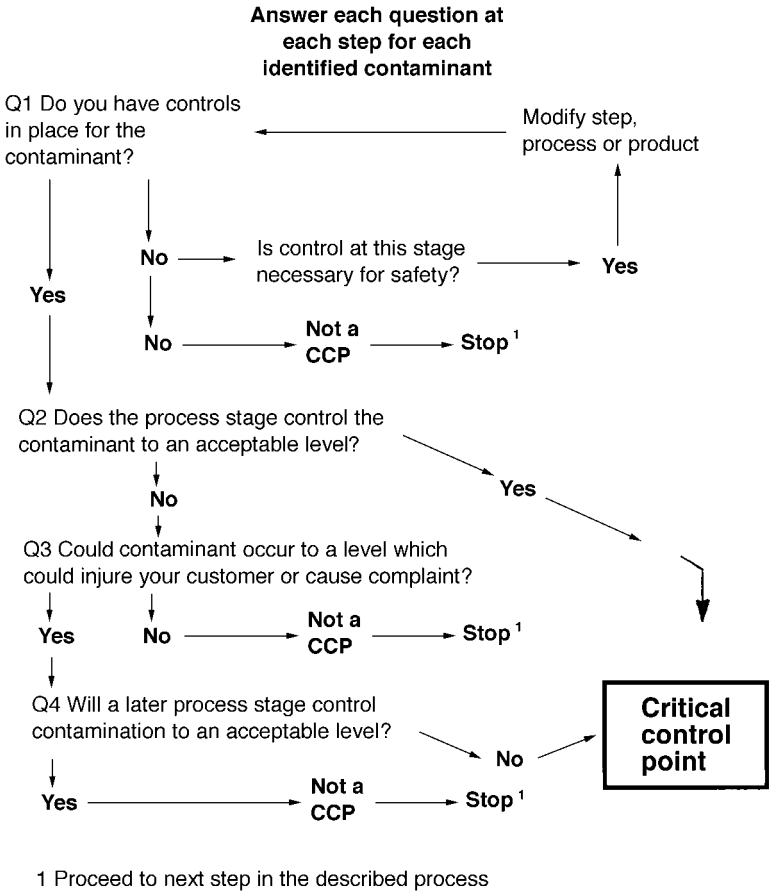


Fig. 1.18 Critical control point decision tree.
(From Fellows *et al.* (1995).)

identify areas of control where failure has not yet been experienced, making it particularly useful for new operations.

Implementation of a HACCP scheme involves the following stages:

1. Produce a detailed flow diagram of the process, including production methods and schedules, preparation and transport of raw materials, in-process stock etc., confirmed by a site visit.
2. Identify the essential characteristics of the product and its use and define the hazards or potential hazards that could threaten the consumer or the product. Included in this is the product formulation, handling procedures and storage conditions, packaging, expected customer handling and the target consumer groups.
3. Consider all stages in the process, including realistic process deviations, and identify critical stages (critical control points (CCPs)) which must be controlled to assure safety. The judgement of risk is made using one of three methods: probabilistic, comparative or pragmatic, and should be made by people who have a high degree of expertise and experience.
4. Devise target levels and critical limits for each CCP and produce effective procedures to monitor them, to verify them and to implement corrective action.

Monitoring may be by physical, microbiological or chemical tests, or by visual or sensory observations. All monitoring procedures should be recorded and they should also include the location of the CCP, the frequency of monitoring and satisfactory compliance criteria. Examples include cleaning procedures (what is cleaned, how and when it is cleaned, who cleans it and what with), temperatures of foods, processing conditions, recipe formulations, hygienic practices, opportunities for cross-contamination and workers' illness or infections.

5. Establish documentation to record the HACCP system and the procedures needed to review it.

Examples of HACCP applications are described by Dillon and Griffith (1996), Mortimer and Wallace (1998) and Fellows *et al*. (1995).

1.8.2 Hurdle technology

In traditionally preserved foods, such as smoked fish or meat, jams and other preserves, there are a combination of factors that ensure microbiological safety and stability of the food, and thus enable it to be preserved. In smoked products for example, this combination includes heat, reduced moisture content and anti-microbial chemicals deposited from the smoke onto the surface of the food. Some smoked foods may also be dipped or soaked in brine or rubbed with salt before smoking, to impregnate the flesh with salt and thus add a further preservative mechanism. Smoked products may also be chilled or packed in modified atmospheres to further extend the shelf life. In jams and other fruit preserves, the combined factors are heat, a high solids content (reduced water activity (Section 1.5)), and high acidity. These preservative factors also strongly influence the sensory characteristics of the product and contribute to important differences in flavour, texture or colour between different products. In vegetable fermentation (Chapter 7), the desired product quality and microbial stability are achieved by a sequence of factors that arise at different stages in the fermentation process: the addition of salt selects the initial microbial population which uses up the available oxygen in the brine. This reduces the redox potential and inhibits the growth of aerobic spoilage micro-organisms and favours the selection of lactic acid bacteria. These then acidify the product and stabilise it. Further treatments may include pasteurisation (Chapter 11) and packaging (Chapter 24) to extend the shelf life and facilitate distribution.

The demand by consumers for high quality foods having 'fresh' or 'natural' characteristics, that require a minimum amount of preparation has led to the development of ready-to-eat and convenience foods that are preserved using mild technologies. The main preservation technique is refrigeration (Chapter 19), but because of the difficulty in maintaining sufficiently low temperatures throughout the production, distribution and storage chain, additional barriers (or 'hurdles') are required to control the growth of spoilage or pathogenic micro-organisms. The concept of combining several factors to preserve foods has been developed by Leistner (1995) and others into the *Hurdle effect* (each factor is a hurdle that micro-organisms must overcome). This in turn has led to the application of Hurdle Technology,[4] where an understanding of the complex interactions of temperature, water activity, pH, chemical preservatives, etc. (Table 1.17) is used to design a series of hurdles that ensure microbiological safety of processed foods. The

4. Also known as 'combined processes', 'combination preservation' or 'combination techniques'.

Table 1.17 Examples of hurdles used to preserve foods

Type of hurdle	Examples	Chapter
Physical hurdles	Aseptic packaging	12, 25
	Electromagnetic energy (microwave, radio frequency, pulsed magnetic fields, high electric fields)	9, 18
	High temperatures (blanching, pasteurisation, sterilisation, evaporation, extrusion, baking, frying)	10–17
	Ionising radiation	8
	Low temperatures (chilling, freezing)	19, 21
	Modified atmospheres	20
	Packaging films (including active packaging, edible coatings)	24
	Photodynamic inactivation	9
	Ultra-high pressures	9
	Ultrasonication	9
	Ultraviolet radiation	9
Physico-chemical hurdles	Carbon dioxide	20
	Ethanol	7
	Lactic acid	7
	Lactoperoxidase	7
	Low pH	
	Low redox potential	
	Low water activity	15–16, 22
	Maillard reaction products	1, 16
	Organic acids	7
	Oxygen	
	Ozone	
	Phenols	
	Phosphates	
	Salt	
	Smoking	
	Sodium nitrite/nitrate	
	Sodium or potassium sulphite	
	Spices and herbs	
	Surface treatment agents	
Microbially derived hurdles	Antibiotics	7
	Bacteriocins	7
	Competitive flora	
	Protective cultures	

Chapter numbers indicate sources of more detailed information.
Adapted from Leistner and Gorris (1995).

hurdles are also used to improve the quality of foods and the economic properties (for example, the weight of water that can be added to a food, consistent with its microbial stability). To be successful, the hurdles must take into account the initial numbers and types of micro-organisms that are likely to be present in the food. The hurdles that are selected should be 'high enough' so that the anticipated numbers of these micro-organisms cannot overcome them. However, the same hurdles that satisfactorily preserve a food when it is properly prepared (Fig. 1.19a), are overcome by a larger initial population of micro-organisms (Fig. 1.19b), when for example raw materials are not adequately cleaned (Chapter 3). In this example, the main hurdles are low water activity and chemical preservatives in the product, with storage temperature, pH and redox potential having a smaller effect. Blanching vegetables or fruits (Chapter 10) has a similar effect in reducing initial numbers of micro-organisms before freezing or drying. If

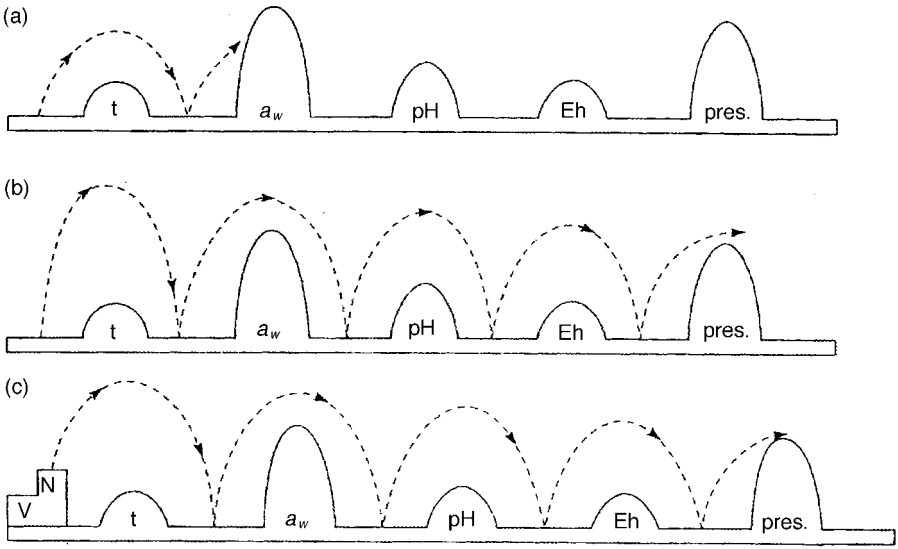


Fig. 1.19 Examples of hurdles in food processing.
(t = chilling, a_w = low water activity, pH = acidification, Eh = low redox potential, pres. = preservatives, V = vitamins, N = nutrients.) (Adapted from Leistner and Gorris (1995).)

in Fig 1.19, the same hurdles are used with a different product that is richer in nutrients that can support microbial growth (Fig. 1.19c), again the hurdles may be inadequate to preserve it and a different combination may be needed or the height of the hurdles increased. It should be noted that although the hurdles in Fig. 1.19 are represented as a sequence, in practice the different factors may operate simultaneously, synergistically or sequentially.

The combination of hurdle technology and HACCP in process design is described by Leistner (1994). By combining hurdles, the intensity of individual preservation techniques can be kept comparatively low to minimise loss of product quality, while overall there is a high impact on controlling microbial growth. Examples of novel mild processes that retain product quality are described in Chapter 9 ('minimal' processes), Chapter 18 (ohmic heating) and Chapters 19 and 20 (sous-vide cooking and modified atmospheres).

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1.10 References

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